1.8.2 DMA: GATE CSE 2005 | Question: 70

Consider a disk drive with the following specifications:

16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, rotation speed 3000 rpm. The disk is operated in cycle stealing mode whereby whenever one 4 byte word is ready it is sent to memory; similarly, for writing, the disk interface reads a 4 byte word from the memory in each DMA cycle. Memory cycle time is 40 nsec. The maximum percentage of time that the CPU gets blocked during DMA operation is:

A. 10 B. 25 C. 40 D. 50

gatecse-2005 co-and-architecture disk normal dma

Answer key

1.8.3 DMA: GATE CSE 2016 | Set 1 | Question: 31

The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of $29,154$ kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is __________.

gatecse-2016-set1 co-and-architecture dma normal numerical-answers

Answer key

1.8.4 DMA: GATE CSE 2021 | Set 2 | Question: 20

Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through *cycle stealing* at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is ___________ bits per second.

gatecse-2021-set2 numerical-answers co-and-architecture dma one-mark

Answer key

1.8.5 DMA: GATE CSE 2022 | Question: 7

Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?

A. DMA based I/O transfer
B. Interrupt driven I/O transfer
C. Polling based I/O transfer
D. Programmed I/O transfer

gatecse-2022 co-and-architecture dma one-mark

Answer key

1.8.6 DMA: GATE CSE 2024 | Set 1 | Question: 5

Which one of the following statements is FALSE?

A. In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
B. For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
C. Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
D. The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

gatecse-2024-set1 co-and-architecture dma one-mark

Answer key

1.8.7 DMA: GATE CSE 2024 | Set 2 | Question: 1

Consider a computer with a 4MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate *(in bits per second)* of the DMA controller if 1% of the processor cycles are used for DMA?

A. $2,56,000$ B. $3,200$ C. $25,60,000$ D. $32,000$

gatecse-2024-set2 co-and-architecture dma one-mark

Answer key

1.8.8 DMA: GATE IT 2004 | Question: 51

The storage area of a disk has the innermost diameter of 10 cm and outermost diameter of 20 cm. The maximum storage density of the disk is 1400 bits/cm. The disk rotates at a speed of 4200 RPM. The main memory of a computer has 64-bit word length and $1\mu s$ cycle time. If cycle stealing is used for data transfer from the disk, the percentage of memory cycles stolen for transferring one word is

A. 0.5% B. 1% C. 5% D. 10%

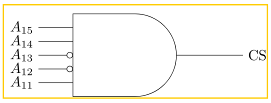
gateit-2004 co-and-architecture dma normal

Answer key

1.9 DRAM (1)

1.9.1 DRAM: GATE CSE 2019 | Question: 2

The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A_{15} to A_0. What is the range of address (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



A. C800 to CFFF B. CA00 to CAFF
C. C800 to C8FF D. DA00 to DFFF

gatecse-2019 co-and-architecture dram memory-interfacing one-mark

Answer key

1.10 Data Dependency (4)

1.10.1 Data Dependency: GATE CSE 2005 | Question: 68

A 5 stage pipelined CPU has the following sequence of stages:

- IF – instruction fetch from instruction memory
- RD – Instruction decode and register read
- EX – Execute: ALU operation for data and address computation
- MA – Data memory access – for write access, the register read at RD state is used.
- WB – Register write back

Consider the following sequence of instructions:

- I_1: $L\ R0, loc\ 1;\ R0 \Leftarrow M[loc1]$
- I_2: $A\ R0, R0;\ R0 \Leftarrow R0 + R0$
- I_3: $S\ R2, R0;\ R2 \Leftarrow R2 - R0$

Let each stage take one clock cycle.

What is the number of clock cycles taken to complete the above sequence of instructions starting from the fetch of I_1?

A. 8 B. 10 C. 12 D. 15

gatecse-2005 co-and-architecture pipelining normal data-dependency

Answer key

1.10.2 Data Dependency: GATE CSE 2008 | Question: 36

Which of the following are NOT true in a pipelined processor?

I. Bypassing can handle all RAW hazards
II. Register renaming can eliminate all register carried WAR hazards
III. Control hazard penalties can be eliminated by dynamic branch prediction

A. I and II only B. I and III only C. II and III only D. I, II and III

gatecse-2008 pipelining co-and-architecture normal data-dependency

Answer key

1.10.3 Data Dependency: GATE CSE 2015 | Set 3 | Question: 47

Consider the following code sequence having five instructions from I_1 to I_5. Each of these instructions has the following format.

OP Ri, Rj, Rk

Where operation OP is performed on contents of registers Rj and Rk and the result is stored in register Ri.

I_1: ADD R1, R2, R3

I_2: MUL R7, R1, R3

I_3: SUB R4, R1, R5

I_4: ADD R3, R2, R4

I_5: MUL R7, R8, R9

Consider the following three statements.

S1: There is an anti-dependence between instructions I_2 and I_5

S2: There is an anti-dependence between instructions I_2 and I_4

S3: Within an instruction pipeline an anti-dependence always creates one or more stalls

Which one of the above statements is/are correct?

A. Only S1 is true
B. Only S2 is true
C. Only S1 and S3 are true
D. Only S2 and S3 are true

gatecse-2015-set3 co-and-architecture pipelining data-dependency normal

Answer key

1.10.4 Data Dependency: GATE IT 2007 | Question: 39

Data forwarding techniques can be used to speed up the operation in presence of data dependencies. Consider the following replacements of LHS with RHS.

$$\begin{array}{ll} \text{i.} & R1 \to Loc, Loc \to R2 \quad \equiv R1 \to R2, R1 \to Loc \\ \text{ii.} & R1 \to Loc, Loc \to R2 \quad \equiv R1 \to R2 \\ \text{iii.} & R1 \to Loc, R2 \to Loc \quad \equiv R1 \to Loc \\ \text{iv.} & R1 \to Loc, R2 \to Loc \quad \equiv R2 \to Loc \end{array}$$

In which of the following options, will the result of executing the RHS be the same as executing the LHS irrespective of the instructions that follow ?

A. i and iii
B. i and iv
C. ii and iii
D. ii and iv

gateit-2007 data-dependency co-and-architecture

Answer key

1.11 Data Hazards (1)

1.11.1 Data Hazards: GATE CSE 2026 | Set 1 | Question: 6

Which one of the following dependencies among the register operands of different instructions can cause a data hazard in a pipelined processor?

A. Read-after-read

B. Read-after-write

C. Write-after-read

D. Write-after-write

gatecse-2026-set1 co-and-architecture data-hazards easy one-mark

Answer key

1.12 Data Path (7)

✍ **Practice Test:** Test 1 (6Q)

1.12.1 Data Path: GATE CSE 1990 | Question: 8a

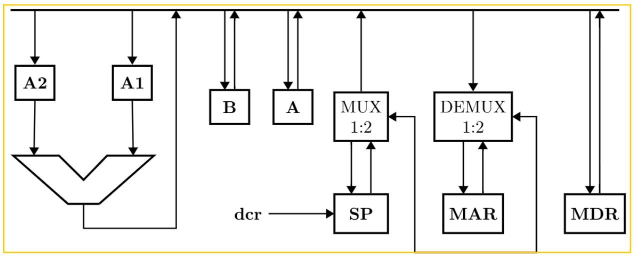
A single bus CPU consists of four general purpose register, namely, $R0, \ldots, R3$, $\mathrm{ALU}, \mathrm{MAR}, \mathrm{MDR}, \mathrm{PC}, \mathrm{SP}$ and IR (Instruction Register). Assuming suitable microinstructions, write a microroutine for the instruction, $\mathrm{ADD}\ R0, R1$.

gate1990 descriptive co-and-architecture data-path

Answer key

1.12.2 Data Path: GATE CSE 2001 | Question: 2.13

Consider the following data path of a simple non-pipelined CPU. The registers A, B, A_1, A_2, MDR, the **bus** and the ALU are $8\text{-}bit$ wide. SP and MAR are $16\text{-}bit$ registers. The MUX is of size $8 \times (2:1)$ and the DEMUX is of size $8 \times (1:2)$. Each memory operation takes 2 CPU clock cycles and uses MAR (Memory Address Register) and MDR (Memory Date Register). SP can be decremented locally.



The CPU instruction "push r" where, $r = A$ or B has the specification

- $M[SP] \leftarrow r$
- $SP \leftarrow SP - 1$

How many CPU clock cycles are required to execute the "push r" instruction?

A. 2

B. 3

C. 4

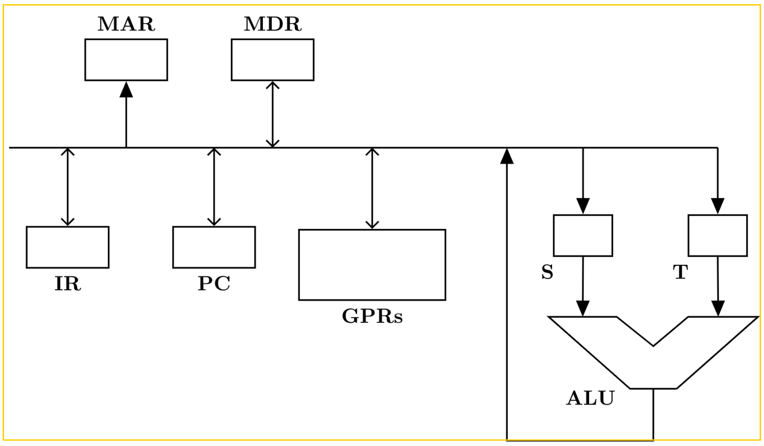
D. 5

gatecse-2001 co-and-architecture data-path machine-instruction normal

Answer key

1.12.3 Data Path: GATE CSE 2005 | Question: 79

Consider the following data path of a CPU.



The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation – the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR.

The instruction "$\mathrm{add\ R0, R1}$" has the register transfer interpretation $\mathrm{R0} \Leftarrow \mathrm{R0} + \mathrm{R1}$. The minimum number of clock cycles needed for execution cycle of this instruction is:

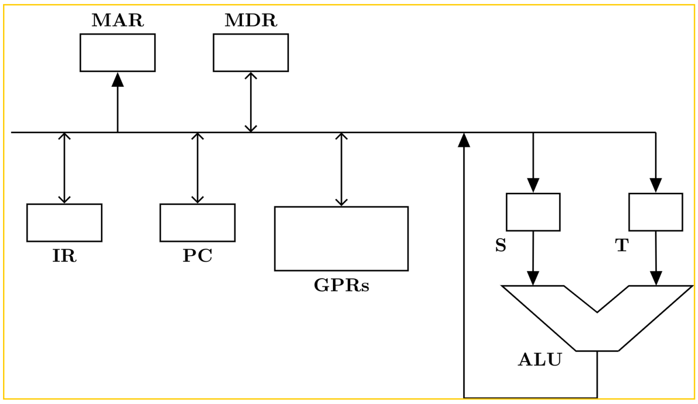
A. 2 B. 3 C. 4 D. 5

gatecse-2005 co-and-architecture machine-instruction data-path normal

Answer key

1.12.4 Data Path: GATE CSE 2005 | Question: 80

Consider the following data path of a CPU.



The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation – the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR.

The instruction "call Rn, sub" is a two word instruction. Assuming that PC is incremented during the fetch cycle of the first word of the instruction, its register transfer interpretation is

$\mathrm{Rn} \leftarrow \mathrm{PC} + 1$;

$\mathrm{PC} \leftarrow \mathrm{M[PC]}$;

The minimum number of **CPU** clock cycles needed during the execution cycle of this instruction is:

A. 2 B. 3 C. 4 D. 5

co-and-architecture normal gatecse-2005 data-path machine-instruction

Answer key

1.12.5 Data Path: GATE CSE 2016 | Set 2 | Question: 30

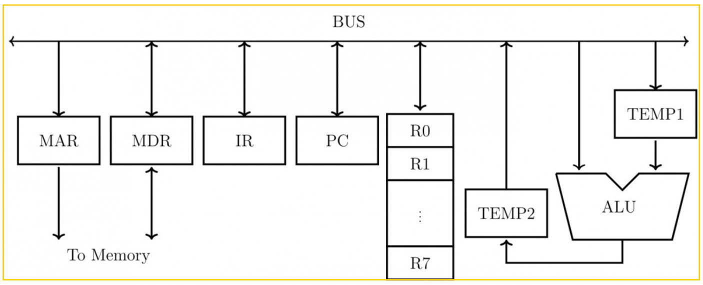
Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G, respectively. Given two instances of U_F and two instances of U_G, it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is ______________ nanoseconds.

gatecse-2016-set2 co-and-architecture data-path normal numerical-answers

Answer key

1.12.6 Data Path: GATE CSE 2020 | Question: 4

Consider the following data path diagram.



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r$, TEMP1_r, ALU_{add}, TEMP2_w
2. $R1_r$, TEMP1_w
3. PC_r, MAR_w, MEM_r
4. TEMP2_r, R0_w
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps?

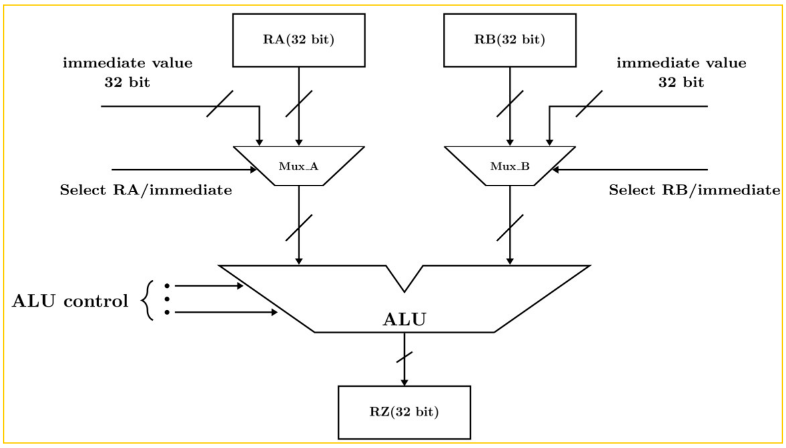
A. $2,1,4,5,3$ B. $1,2,4,3,5$ C. $3,5,2,1,4$ D. $3,5,1,2,4$

gatecse-2020 co-and-architecture data-path one-mark

Answer key

1.12.7 Data Path: GATE CSE 2025 | Set 1 | Question: 17

A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?



A. The data path can implement arithmetic operations involving two registers.
B. The data path can implement arithmetic operations involving one register and one immediate value.

C. The data path can implement arithmetic operations involving two immediate values.
D. The data path can only implement arithmetic operations involving one register and one immediate value.

gatecse2025-set1 co-and-architecture data-path multiple-selects one-mark

Answer key

1.13 Direct Mapping (5)

1.13.1 Direct Mapping: GATE CSE 2022 | Question: 44

Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words $P, Q, R,$ and S are accessed in that order 10 times (i.e., $PQRSPQRS\ldots$). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of $P, Q, R,$ and S are $0xA248, 0xC28A, 0xCA8A,$ and $0xA262$, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

A. Every access to S is a hit.
B. Once P is brought to the cache it is never evicted.
C. At the end of the execution only R and S reside in the cache.
D. Every access to R evicts Q from the cache.

gatecse-2022 co-and-architecture direct-mapping multiple-selects two-marks cache-memory

Answer key

1.13.2 Direct Mapping: GATE CSE 2025 | Set 1 | Question: 26

Consider a memory system with $1M$ bytes of main memory and 16 K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the tag values? [Assume memory is byte addressable, $1\text{ K} = 2^{10}$, $1\text{M} = 2^{20}$.]

A. 6×2^{10} B. 8×2^{10} C. 2^{12} D. 2^{14}

gatecse2025-set1 co-and-architecture cache-memory direct-mapping easy two-marks

Answer key

1.13.3 Direct Mapping: GATE CSE 2025 | Set 2 | Question: 29

For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block.

Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?

A. 64 KB and 4 KB B. 128 KB and 16 KB C. 64 KB and 8 KB D. 128 KB and 6 KB

gatecse2025-set2 co-and-architecture direct-mapping cache-memory two-marks

Answer key

1.13.4 Direct Mapping: GATE CSE 2026 | Set 2 | Question: 42

Consider a system with a processor and a 4 KB direct mapped cache with block size of 16 bytes. The system has a 16 MB physical memory. Four words $P, Q, R,$ and S are accessed by the processor in the same order 10 times. That is, there are a total of 40 memory references in the sequence $P, Q, R, S, P, Q, R, S, \ldots$.

Assume that the cache memory is initially empty. The physical addresses of the words are given below (1 word $= 1$ byte).

P: 0x845B32, Q: 0x845B26, R: 0x845B36, S: 0x846B32

Which of the following statements is/are true?
Note: $1\text{ K} = 2^{10}$ and $1\text{M} = 2^{20}$

A. Every access to P results in a cache miss
B. Every access to R results in a cache hit
C. Every access to Q results in a cache miss
D. Except the first access to S, all subsequent accesses to S result in cache hits

gatecse-2026-set2 co-and-architecture direct-mapping cache-memory multiple-selects two-marks

Answer key

1.13.5 Direct Mapping: GATE CSE 2026 | Set 2 | Question: 46

Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address 0xA2C28 is mapped to the cache block number 176_{10}. The maximum possible size of the cache (in KB) for this configuration is ________. (answer in integer)

Note: $1\text{ K} = 2^{10}$ and $1\text{M} = 2^{20}$

gatecse-2026-set2 co-and-architecture direct-mapping cache-memory numerical-answers two-marks

Answer key

1.14 Disk (1)

1.14.1 Disk: GATE CSE 2026 | Set 1 | Question: 49

Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400. The sector size is 1024 bytes. It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: $5, 12$ and 7.

The total time for the data transfer (in milliseconds) from the hard disk is ________. (rounded off to one decimal place)

gatecse-2026-set1 numerical-answers operating-system two-marks disk

Answer key

1.15 Hazards (1)

1.15.1 Hazards: GATE CSE 2026 | Set 1 | Question: 50

The EX stage of a pipelined processor performs the memory read operations for LOAD instructions, and the operations for the arithmetic and logic instructions. Let t_{EX} denote the time taken by the EX stage to perform the operation for an instruction. For each instruction type, the values of t_{EX} and M (the number of instructions of that type in a sequence of 100 instructions for a program P), are given in the table below.

The duration of the pipeline clock cycle is 1 nanosecond. Assume that the latch time for the interstage buffers in the pipeline is negligible.

Instruction	t_{EX} in nanoseconds	M
LOAD	1.8	15
IMUL	1.5	10
IDIV	2.5	5
FADD	1.7	10
FSUB	1.7	5
FMUL	2.8	15
FDIV	3.2	5
All other instructions	Less than 1.0	35

When program P is executed, the number of clock cycles for which the pipeline is stalled due to structural hazards in the EX stage is ____. (answer in integer)

gatecse-2026-set1 numerical-answers co-and-architecture pipelining hazards two-marks

Answer key

1.16 IO Handling (8)

1.16.1 IO Handling: GATE CSE 1987 | Question: 2a

State whether the following statements are TRUE or FALSE

In a microprocessor-based system, if a bus (DMA) request and an interrupt request arrive simultaneously, the microprocessor attends first to the bus request.

gate1987 co-and-architecture interrupts io-handling true-false

Answer key

1.16.2 IO Handling: GATE CSE 1987 | Question: 2b

State whether the following statements are TRUE or FALSE:

Data transfer between a microprocessor and an I/O device is usually faster in memory-mapped-I/O scheme than in I/O-mapped -I/O scheme.

gate1987 co-and-architecture io-handling true-false

Answer key

1.16.3 IO Handling: GATE CSE 1990 | Question: 4-ii

State whether the following statements are TRUE or FALSE with reason:

The data transfer between memory and I/O devices using programmed I/O is faster than interrupt-driven I/O.

gate1990 true-false co-and-architecture io-handling interrupts

Answer key

1.16.4 IO Handling: GATE CSE 1996 | Question: 1.24

For the daisy chain scheme of connecting I/O devices, which of the following statements is true?

A. It gives non-uniform priority to various devices
B. It gives uniform priority to all devices
C. It is only useful for connecting slow devices to a processor device
D. It requires a separate interrupt pin on the processor for each device

gate1996 co-and-architecture io-handling normal

Answer key

1.16.5 IO Handling: GATE CSE 1996 | Question: 25

A hard disk is connected to a 50 MHz processor through a DMA controller. Assume that the initial set-up of a DMA transfer takes 1000 clock cycles for the processor, and assume that the handling of the interrupt at DMA completion requires 500 clock cycles for the processor. The hard disk has a transfer rate of 2000 Kbytes/sec and average block transferred is 4 K bytes. What fraction of the processor time is consumed by the disk, if the disk is actively transferring 100% of the time?

gate1996 co-and-architecture io-handling dma numerical-answers normal

Answer key

1.16.6 IO Handling: GATE CSE 1997 | Question: 2.4

The correct matching for the following pairs is:

(A) DMA I/O	(1) High speed RAM
(B) Cache	(2) Disk
(C) Interrupt I/O	(3) Printer
(D) Condition Code Register	(4) ALU

A. $A-4$ $B-3$ $C-1$ $D-2$

B. $A-2$ $B-1$ $C-3$ $D-4$

C. $A-4$ $B-3$ $C-2$ $D-1$

D. $A-2$ $B-3$ $C-4$ $D-1$

gate1997 co-and-architecture normal io-handling match-the-following

Answer key

1.16.7 IO Handling: GATE CSE 2008 | Question: 64, ISRO2009-13

Which of the following statements about synchronous and asynchronous I/O is NOT true?

A. An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O

B. In both synchronous and asynchronous I/O, an ISR (Interrupt Service Routine) is invoked after completion of the I/O

C. A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O

D. In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

gatecse-2008 operating-system io-handling normal isro2009

Answer key

1.16.8 IO Handling: GATE CSE 2011 | Question: 28

On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```
    Initialize the address register
    Initialize the count to 500
LOOP:   Load a byte from device
    Store in memory at address given by address register
    Increment the address register
    Decrement the count
    If count !=0 go to LOOP
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

A. 3.4 B. 4.4 C. 5.1 D. 6.7

gatecse-2011 co-and-architecture dma normal io-handling

Answer key

1.17 Instruction Execution (7)

Practice Test: Test 1 (7Q)

1.17.1 Instruction Execution: GATE CSE 1990 | Question: 4-iii

State whether the following statements are TRUE or FALSE with reason:

The flags are affected when conditional CALL or JUMP instructions are executed.

gate1990 true-false co-and-architecture instruction-execution

Answer key

1.17.2 Instruction Execution: GATE CSE 1992 | Question: 01-iv

Many of the advanced microprocessors prefetch instructions and store it in an instruction buffer to speed up processing. This speed up is achieved because _________

gate1992 co-and-architecture easy instruction-execution fill-in-the-blanks

Answer key

1.17.3 Instruction Execution: GATE CSE 1995 | Question: 1.2

Which of the following statements is true?

A. ROM is a Read/Write memory
B. PC points to the last instruction that was executed
C. Stack works on the principle of LIFO
D. All instructions affect the flags

gate1995 co-and-architecture normal instruction-execution

Answer key

1.17.4 Instruction Execution: GATE CSE 2002 | Question: 1.13

Which of the following is not a form of memory

A. instruction cache
B. instruction register
C. instruction opcode
D. translation look-a-side buffer

gatecse-2002 co-and-architecture easy instruction-execution

Answer key

1.17.5 Instruction Execution: GATE CSE 2006 | Question: 43

Consider a new instruction named branch-on-bit-set (mnemonic bbs). The instruction “bbs reg, pos, label” jumps to label if bit in position pos of register operand reg is one. A register is 32 -bits wide and the bits are numbered 0 to 31, bit in position 0 being the least significant. Consider the following emulation of this instruction on a processor that does not have bbs implemented.

$$temp \leftarrow reg\&mask$$

Branch to label if temp is non-zero. The variable temp is a temporary register. For correct emulation, the variable mask must be generated by

A. $mask \leftarrow 0x1 << pos$
B. $mask \leftarrow 0xffffffff << pos$
C. $mask \leftarrow pos$
D. $mask \leftarrow 0xf$

gatecse-2006 co-and-architecture normal instruction-execution

Answer key

1.17.6 Instruction Execution: GATE CSE 2017 | Set 1 | Question: 49

Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence

Instr. No.	Instruction
i:	add R2, R3, R4
i+1:	sub R5, R6, R7
i+2:	cmp R1, R9, R10
i+3:	beq R1, Offset

If the target of the branch instruction is i, then the decimal value of the Offset is ____________ .

gatecse-2017-set1 co-and-architecture normal numerical-answers instruction-execution

Answer key

1.17.7 Instruction Execution: GATE CSE 2025 | Set 2 | Question: 51

An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is ___________. (rounded off to one decimal place)

Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed
Branch	2	2.25×10^8
Load	5	1.20×10^8
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

gatecse2025-set2 co-and-architecture instruction-execution clock-cycles numerical-answers two-marks

Answer key

1.18 Instruction Format (11)

Practice Test: Test 1 (10Q)

1.18.1 Instruction Format: GATE CSE 1988 | Question: 2-ii

Using an expanding opcode encoding for instructions, is it possible to encode all of the following in an instruction format shown in the below figure. Justify your answer.

14 double address instructions
127 single address instructions
60 no address (zero address) instructions

← 4 bits →	← 6 bits →	← 6 bits →
Opcode	Operand 1 Address	Operand 2 Address

gate1988 normal co-and-architecture instruction-format descriptive

Answer key

1.18.2 Instruction Format: GATE CSE 1992 | Question: 01-vi

In an 11-bit computer instruction format, the size of address field is 4-bits. The computer uses expanding OP code technique and has 5 two-address instructions and 32 one-address instructions. The number of zero-address instructions it can support is ________

gate1992 co-and-architecture machine-instruction instruction-format normal numerical-answers

Answer key

1.18.3 Instruction Format: GATE CSE 1994 | Question: 3.2

State True or False with one line explanation

Expanding opcode instruction formats are commonly employed in RISC. (Reduced Instruction Set Computers) machines.

gate1994 co-and-architecture machine-instruction instruction-format normal true-false

Answer key

1.18.4 Instruction Format: GATE CSE 2014 | Set 1 | Question: 9

A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is ____________

gatecse-2014-set1 co-and-architecture machine-instruction instruction-format numerical-answers normal

Answer key

1.18.5 Instruction Format: GATE CSE 2016 | Set 2 | Question: 31

Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _________.

gatecse-2016-set2 instruction-format machine-instruction co-and-architecture normal numerical-answers

Answer key

1.18.6 Instruction Format: GATE CSE 2018 | Question: 51

A processor has 16 integer registers $(R0, R1, \ldots, R15)$ and 64 floating point registers $(F0, F1, \ldots, F63)$. It uses a 2- byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands $(3Rs)$. Type-2 category consists of eight instructions, each with 2 floating point register operands $(2Fs)$. Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand $(1R+1F)$. Type-4 category consists of N instructions, each with a floating point register operand $(1F)$.

The maximum value of N is _________.

gatecse-2018 co-and-architecture machine-instruction instruction-format numerical-answers two-marks

Answer key

1.18.7 Instruction Format: GATE CSE 2020 | Question: 44

A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _______.

gatecse-2020 co-and-architecture numerical-answers instruction-format machine-instruction two-marks

Answer key

1.18.8 Instruction Format: GATE CSE 2024 | Set 2 | Question: 47

A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is ___________.

gatecse-2024-set2 numerical-answers co-and-architecture instruction-format two-marks

Answer key

1.18.9 Instruction Format: GATE CSE 2024 | Set 2 | Question: 51

A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.

R - type Instruction Format:

$OPCODE$	$UNUSED$	$DSTRegister$	$SRCRegister1$	$SRCRegister2$

I - type Instruction Format:

$OPCODE$	$DSTRegister$	$SRCRegister$	$\#Immediatevalue/address$

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.

Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field. The value of $\mathrm{X+2Y+Z}$ is ___________.

gatecse-2024-set2 numerical-answers co-and-architecture instruction-format two-marks

Answer key

1.18.10 Instruction Format: GATE CSE 2026 | Set 1 | Question: 5

Consider a processor P whose instruction set architecture is the load-store architecture. The instruction format is such that the first operand of any instruction is the destination operand.

Which one of the following sequences of instructions corresponds to the high-level language statement $\mathrm{Z = X + Y}$?

Note: X, Y, and Z are memory operands. $\mathrm{R0}$, $\mathrm{R1}$, and R2 are registers.

A. ADD Z, X, Y
B. LOAD $\mathrm{R0}$, X ADD Z, $\mathrm{R0}$, Y
C. ADD $\mathrm{R0}$, X, Y STORE Z, $\mathrm{R0}$
D. LOAD $\mathrm{R0}$, X LOAD $\mathrm{R1}$, Y ADD R 2, R0, R1 STORE Z, R2

gatecse-2026-set1 co-and-architecture instruction-format one-mark

Answer key

1.18.11 Instruction Format: GATE CSE 2026 | Set 2 | Question: 34

Consider a processor that has 16 general purpose registers and it uses 2-byte instruction format for all its instructions. Variable-sized opcodes are permitted. There are three different types of instructions; M-type, R-type, and C-type. Each M-type instruction has 2 register operands and a 6-bit immediate operand. Each Rtype instruction has 3 register operands. Each C-type instruction has a register operand and a 6 -bit offset value. If there are 2 unique M-type opcodes and 7 unique R-type opcodes, which one of the following options gives the maximum number of unique opcodes possible for C-type instructions?

A. 8 B. 4 C. 64 D. 16

gatecse-2026-set2 co-and-architecture machine-instruction instruction-format two-marks

Answer key

1.19 Instruction Set Architecture (1)

1.19.1 Instruction Set Architecture: GATE CSE 2025 | Set 2 | Question: 18

Which of the following is/are part of an Instruction Set Architecture of a processor?

A. The size of the cache memory
B. The clock frequency of the processor
C. The number of cache memory levels
D. The total number of registers

gatecse2025-set2 co-and-architecture instruction-set-architecture multiple-selects easy one-mark

Answer key

1.20 Interrupts (10)

Practice Test: Test 1 (11Q)

1.20.1 Interrupts: GATE CSE 1987 | Question: 1-viii

On receiving an interrupt from a I/O device the CPU:

A. Halts for a predetermined time.
B. Hands over control of address bus and data bus to the interrupting device.
C. Branches off to the interrupt service routine immediately.
D. Branches off to the interrupt service routine after completion of the current instruction.

gate1987 co-and-architecture interrupts

Answer key

1.20.2 Interrupts: GATE CSE 1995 | Question: 1.3

In a vectored interrupt:

A. The branch address is assigned to a fixed location in memory
B. The interrupting source supplies the branch information to the processor through an interrupt vector
C. The branch address is obtained from a register in the processor
D. None of the above

gate1995 co-and-architecture interrupts normal

Answer key

1.20.3 Interrupts: GATE CSE 1998 | Question: 1.20

Which of the following is true?

A. Unless enabled, a CPU will not be able to process interrupts.
B. Loop instructions cannot be interrupted till they complete.
C. A processor checks for interrupts before executing a new instruction.
D. Only level triggered interrupts are possible on microprocessors.

gate1998 co-and-architecture interrupts normal

Answer key

1.20.4 Interrupts: GATE CSE 2002 | Question: 1.9

A device employing INTR line for device interrupt puts the CALL instruction on the data bus while:

A. INTA is active
B. HOLD is active
C. READY is inactive
D. None of the above

gatecse-2002 co-and-architecture interrupts normal

Answer key

1.20.5 Interrupts: GATE CSE 2005 | Question: 69

A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4μsec. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode?

A. 15
B. 25
C. 35
D. 45

gatecse-2005 co-and-architecture interrupts

Answer key

1.20.6 Interrupts: GATE CSE 2009 | Question: 8, UGCNET-June2012-III: 58

A CPU generally handles an interrupt by executing an interrupt service routine:

A. As soon as an interrupt is raised.
B. By checking the interrupt register at the end of fetch cycle.
C. By checking the interrupt register after finishing the execution of the current instruction.
D. By checking the interrupt register at fixed time intervals.

gatecse-2009 co-and-architecture interrupts normal ugcnetcse-june2012-paper3

Answer key

1.20.7 Interrupts: GATE CSE 2020 | Question: 3

Consider the following statements.

I. Daisy chaining is used to assign priorities in attending interrupts.
II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt.
III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.
IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.

Which of the above statements is/are TRUE?

A. I and II only
B. I and IV only
C. I and III only
D. III only

gatecse-2020 co-and-architecture interrupts one-mark

Answer key

1.20.8 Interrupts: GATE CSE 2023 | Question: 24

A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes 100 μs (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional 200 μs to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke.

In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke.

The ratio $\frac{T_1}{T_2}$ is ______________. (Rounded off to one decimal place)

gatecse-2023 co-and-architecture interrupts numerical-answers one-mark

Answer key

1.20.9 Interrupts: GATE CSE 2025 | Set 1 | Question: 1

Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives:

i. The processor saves the content of the program counter.
ii. The program counter is loaded with the start address of the ISR.
iii. The processor finishes the present instruction.

Which ONE of the following is the CORRECT sequence of steps?

A. (iii), (i), (ii)
B. (i), (iii), (ii)
C. (i), (ii), (iii)
D. (iii), (ii), (i)

gatecse2025-set1 co-and-architecture interrupts easy one-mark

Answer key

1.20.10 Interrupts: GATE CSE 2026 | Set 2 | Question: 8

Consider the following two statements about interrupt handling mechanisms in a CPU.

$S1$: In non-vectored interrupt mechanism, it usually takes more time to start the Interrupt Service Routine (ISR) when compared to that in a vectored interrupt mechanism.

$S2$: In daisy-chain interrupt mechanism, the CPU polls all the input devices individually to determine the source of the interrupt.

Which one of the following options is correct with respect to $S1$ and $S2$?

A. Both $S1$ and $S2$ are true
B. Both $S1$ and $S2$ are false
C. $S1$ is true and $S2$ is false
D. $S1$ is false and $S2$ is true

gatecse-2026-set2 co-and-architecture interrupts one-mark

Answer key

1.21 Machine Instruction (21)

Practice Tests: Test 1 (15Q) Test 2 (8Q)

1.21.1 Machine Instruction: GATE CSE 1988 | Question: 9i

The following program fragment was written in an assembly language for a single address computer with one accumulator register:

```
LOAD B
MULT C
STORE T1
ADD A
STORE T2
MULT T2
ADD T1
STORE Z
```

Give the arithmetic expression implemented by the fragment.

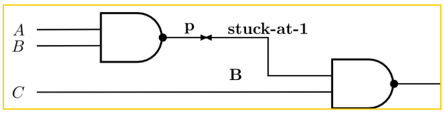
gate1988 normal descriptive co-and-architecture machine-instruction

Answer key

1.21.2 Machine Instruction: GATE CSE 1994 | Question: 12

A. Assume that a CPU has only two registers R_1 and R_2 and that only the following instruction is available $XOR\,R_i, R_j; \{R_j \leftarrow R_i \oplus R_j$, for $i, j = 1, 2\}$
Using this XOR instruction, find an instruction sequence in order to exchange the contents of the registers R_1 and R_2

B. The line p of the circuit shown in figure has stuck at 1 fault. Determine an input test to detect the fault.



gate1994 co-and-architecture machine-instruction normal descriptive

Answer key

1.21.3 Machine Instruction: GATE CSE 1999 | Question: 17

Consider the following program fragment in the assembly language of a certain hypothetical processor. The processor has three general purpose registers $R1$, $R2$ and $R3$. The meanings of the instructions are shown by comments (starting with ;) after the instructions.

```
X:  CMP R1, 0;  Compare R1 and 0, set flags appropriately in status register
    JZ  Z;  Jump if zero to target Z
    MOV R2, R1; Copy contents of R1 to R2
    SHR R1; Shift right R1 by 1 bit
    SHL R1; Shift left R1 by 1 bit
    CMP R2, R1; Compare R2 and R1 and set flag in status register
    JZ  Y;  Jump if zero to target Y
    INC R3; Increment R3 by 1;
Y:  SHR R1; Shift right R1 by 1 bit
    JMP X;  Jump to target X
Z:...
```

A. Initially $R1$, $R2$ and $R3$ contain the values 5,0 and 0 respectively, what are the final values of $R1$ and $R3$ when control reaches Z?

B. In general, if $R1$, $R2$ and $R3$ initially contain the values n, 0, and 0 respectively. What is the final value of $R3$ when control reaches Z?

gate1999 co-and-architecture machine-instruction normal descriptive

Answer key

1.21.4 Machine Instruction: GATE CSE 2003 | Question: 48

Consider the following assembly language program for a hypothetical processor $A, B,$ and C are $8-$bit registers. The meanings of various instructions are shown as comments.

MOV B, #0 ; $B \leftarrow 0$

MOV C, #8 ; $C \leftarrow 8$

Z: CMP C, #0 ; compare C with 0

JZ X ; jump to X if zero flag is set

SUB C, #1 ; $C \leftarrow C - 1$

RRC A, #1 ; right rotate A through carry by one bit. Thus:

; If the initial values of A and the carry flag are $a_7 \dots a_0$ and

; c_0 respectively, their values after the execution of this

; instruction will be $c_0 a_7 \dots a_1$ and a_0 respectively.

JC Y ; jump to Y if carry flag is set

JMP Z ; jump to Z

Y: ADD B, #1 ; $B \leftarrow B + 1$

JMP Z ; jump to Z

X: ;

If the initial value of register A is A0 the value of register B after the program execution will be

A. the number of 0 bits in A_0

B. the number of 1 bits in A_0

C. A_0

D. 8

gatecse-2003 co-and-architecture machine-instruction normal

Answer key

1.21.5 Machine Instruction: GATE CSE 2003 | Question: 49

Consider the following assembly language program for a hypothetical processor A, B, and C are 8 bit registers. The meanings of various instructions are shown as comments.

MOV B, #0 ; $B \leftarrow 0$

MOV C, #8 ; $C \leftarrow 8$

Z: CMP C, #0 ; compare C with 0

JZ X ; jump to X if zero flag is set

SUB C, #1 ; $C \leftarrow C - 1$

RRC A, #1 ; right rotate A through carry by one bit. Thus:

; If the initial values of A and the carry flag are $a_7 .. a_0$ and

; c_0 respectively, their values after the execution of this

; instruction will be $c_0 a_7 .. a_1$ and a_0 respectively.

JC Y ; jump to Y if carry flag is set

JMP Z ; jump to Z

Y: ADD B, #1 ; $B \leftarrow B + 1$

JMP Z ; jump to Z

X:

Which of the following instructions when inserted at location X will ensure that the value of the register A after program execution is as same as its initial value?

A. RRC A,#1

B. NOP; no operation

C. LRC A,#1; left rotate A through carry flag by one bit

D. ADD A,#1

gatecse-2003 co-and-architecture machine-instruction normal

Answer key

1.21.6 Machine Instruction: GATE CSE 2004 | Question: 63

Consider the following program segment for a hypothetical CPU having three user registers R_1, R_2 and R_3.

Instruction	Operation	Instruction size (in words)
MOV $R_1, 5000$	$R_1 \leftarrow$ Memory[5000]	2
MOV $R2, (R1)$	$R2 \leftarrow$ Memory[(R_1)]	1
ADD R_2, R_3	$R2 \leftarrow R_2 + R_3$	1
MOV $6000, R_2$	Memory[6000] $\leftarrow R_2$	2
HALT	Machine Halts	1

Consider that the memory is byte addressable with size 32 bits, and the program has been loaded starting from memory location 1000 (decimal). If an interrupt occurs while the CPU has been halted after executing the HALT instruction, the return address (in decimal) saved in the stack will be

A. 1007 B. 1020 C. 1024 D. 1028

gatecse-2004 co-and-architecture machine-instruction normal

Answer key

1.21.7 Machine Instruction: GATE CSE 2004 | Question: 64

Consider the following program segment for a hypothetical CPU having three user registers R_1, R_2 and R_3.

Instruction	Operation	Instruction size (in Words)
MOV $R_1, 5000$	$R1 \leftarrow$ Memory[5000]	2
MOV $R_2(R_1)$	$R2 \leftarrow$ Memory[(R_1)]	1
ADD R_2, R_3	$R_2 \leftarrow R_2 + R_3$	1
MOV $6000, R_2$	Memory[6000] $\leftarrow R_2$	2
Halt	Machine Halts	1

Let the clock cycles required for various operations be as follows:

Register to/from memory transfer	3 clock cycles
ADD with both operands in register	1 clock cycles
Instruction fetch and decode	2 clock cycles

The total number of clock cycles required to execute the program is

A. 29 B. 24 C. 23 D. 20

gatecse-2004 co-and-architecture machine-instruction normal

Answer key

1.21.8 Machine Instruction: GATE CSE 2006 | Question: 09, ISRO2009-35

A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

A. 400 B. 500 C. 600 D. 700

gatecse-2006 co-and-architecture machine-instruction easy isro2009

Answer key

1.21.9 Machine Instruction: GATE CSE 2007 | Question: 54

In a simplified computer the instructions are:

OP R_j, R_i	Perform R_j OP R_i and store the result in register R_j
OP m, R_i	Perform val OP R_i and store the result in register R_i val denotes the content of the memory location m
MOV m, R_i	Moves the content of memory location m to register R_i
MOV R_i, m	Moves the content of register R_i to memory location m

The computer has only two registers, and OP is either ADD or SUB. Consider the following basic block:

- $t_1 = a + b$
- $t_2 = c + d$
- $t_3 = e - t_2$
- $t_4 = t_1 - t_3$

Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

A. 2 B. 3 C. 5 D. 6

gatecse-2007 co-and-architecture machine-instruction normal

Answer key

1.21.10 Machine Instruction: GATE CSE 2007 | Question: 71

Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1← M[3000]	2
LOOP:	MOV R2,(R3)	R2← M[R3]	1
	ADD R2,R1	R2← R1 + R2	1
	MOV (R3),R2	M[R3]← R2	1
	INC R3	R3← R3 + 1	1
	DEC R1	R1← R1 − 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. The number of memory references for accessing the data in executing the program completely is

A. 10 B. 11 C. 20 D. 21

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.11 Machine Instruction: GATE CSE 2007 | Question: 72

Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1← M[3000]	2
LOOP:	MOV R2,(R3)	R2← M[R3]	1
	ADD R2,R1	R2← R1 + R2	1
	MOV (R3),R2	M[R3]← R2	1
	INC R3	R3← R3 + 1	1
	DEC R1	R1← R1 − 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

A. 100 B. 101 C. 102 D. 110

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.12 Machine Instruction: GATE CSE 2007 | Question: 73

Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	R1← M[3000]	2
LOOP:	MOV R2,(R3)	R2← M[R3]	1
	ADD R2,R1	R2← R1 + R2	1
	MOV (R3),R2	M[R3]← R2	1
	INC R3	R3← R3 + 1	1
	DEC R1	R1← R1 − 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?

A. 1005 B. 1020 C. 1024 D. 1040

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.13 Machine Instruction: GATE CSE 2008 | Question: 34

Which of the following must be true for the RFE (Return From Exception) instruction on a general purpose processor?

I. It must be a trap instruction
II. It must be a privileged instruction
III. An exception cannot be allowed to occur during execution of an RFE instruction

A. I only B. II only C. I and II only D. I, II and III only

gatecse-2008 co-and-architecture machine-instruction normal

Answer key

1.21.14 Machine Instruction: GATE CSE 2015 | Set 2 | Question: 42

Consider a processor with byte-addressable memory. Assume that all registers, including program counter (PC) and Program Status Word (PSW), are size of two bytes. A stack in the main memory is implemented from memory location $(0100)_{16}$ and it grows upward. The stack pointer (SP) points to the top element of the stack. The current value of SP is $(016E)_{16}$. The CALL instruction is of two words, the first word is the op-code and the second word is the starting address of the subroutine (one word = 2 bytes). The CALL instruction is implemented as follows:

- Store the current value of PC in the stack
- Store the value of PSW register in the stack
- Load the statring address of the subroutine in PC

The content of PC just before the fetch of a CALL instruction is $(5FA0)_{16}$. After execution of the CALL instruction, the value of the stack pointer is:

A. $(016A)_{16}$ B. $(016C)_{16}$ C. $(0170)_{16}$ D. $(0172)_{16}$

gatecse-2015-set2 co-and-architecture machine-instruction easy

Answer key

1.21.15 Machine Instruction: GATE CSE 2016 | Set 2 | Question: 10

A processor has 40 distinct instruction and 24 general purpose registers. A 32-bit instruction word has an opcode, two registers operands and an immediate operand. The number of bits available for the immediate operand field is________.

gatecse-2016-set2 machine-instruction co-and-architecture easy numerical-answers

Answer key

1.21.16 Machine Instruction: GATE CSE 2021 | Set 1 | Question: 55

Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY[X] denotes the content at the memory location X.

Instruction	**Semantics**	**Instruction Size** (bytes)
MOV R1, (5000)	$\text{R1} \leftarrow \text{MEMORY[5000]}$	4
MOV R2, (R3)	$\text{R2} \leftarrow \text{MEMORY[R3]}$	4
ADD R2, R1	$\text{R2} \leftarrow \text{R1} + \text{R2}$	2
MOV (R3), R2	$\text{MEMORY[R3]} \leftarrow \text{R2}$	4
INC R3	$\text{R3} \leftarrow \text{R3} + 1$	2
DEC R1	$\text{R1} \leftarrow \text{R1} - 1$	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3020 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is ____________

gatecse-2021-set1 co-and-architecture machine-instruction numerical-answers two-marks

Answer key

1.21.17 Machine Instruction: GATE CSE 2023 | Question: 31

Consider the given C-code and its corresponding assembly code, with a few operands U1-U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable.

```
//C-code

int a[10], b[10], i;
 // int is 32 bit
 for(i=0; i<10; i++)
   a[i] = b[i] * 8;
```

```
;assembly code (; indicates comments)
;r1-r5 are 32-bit integer registers
;initialize r1=0, r2=10
;initialize r3, r4 with base address of a, b

 L01: jeq r1, r2, end  ;if(r1==r2) goto end
 L02: lw, r5, 0(r4)    ;r5 <- Memory[r4+0]
 L03: shl, r5, r5, U1  ;r5 <- r5 << U1
 L04: sw, r5, 0(r3)    ;Memory[r3+0] <- r5
 L05: add, r3, r3, U2  ;r3 <- r3+U2
 L06: add, r4, r4, U3
 L07: add, r1, r1, 1
 L08: jmp U4           ;goto U4
 L09: end
```

Which one of the following options is a CORRECT replacement for operands in the position $(U1, U2, U3, U4)$ in the above assembly code?

A. $(8,4,1,L02)$ B. $(3,4,4,L01)$ C. $(8,1,1,L02)$ D. $(3,1,1,L01)$

gatecse-2023 co-and-architecture machine-instruction two-marks

Answer key

1.21.18 Machine Instruction: GATE CSE 2025 | Set 1 | Question: 27

A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction?

$$\text{ADD R1, \#25 // R1=R1+25}$$

A. 16 B. 20 C. 22 D. 24

gatecse2025-set1 co-and-architecture machine-instruction easy two-marks

Answer key

1.21.19 Machine Instruction: GATE IT 2004 | Question: 46

If we use internal data forwarding to speed up the performance of a CPU (R1, R2 and R3 are registers and M[100] is a memory reference), then the sequence of operations

R1 → M[100]
M[100] → R2
M[100] → R3

can be replaced by

A. R1 → R3
R2 → M[100]

B. M[100] → R2
R1 → R2
R1 → R3

C. R1 → M[100]
R2 → R3

D. R1 → R2
R1 → R3
R1 → M[100]

gateit-2004 co-and-architecture machine-instruction easy

Answer key

1.21.20 Machine Instruction: GATE IT 2007 | Question: 41

Following table indicates the latencies of operations between the instruction producing the result and instruction using the result.

Instruction producing the result	Instruction using the result	Latency
ALU Operation	ALU Operation	2
ALU Operation	Store	2
Load	ALU Operation	1
Load	Store	0

Consider the following code segment:

```
Load R1, Loc 1; Load R1 from memory location Loc1
Load R2, Loc 2; Load R2 from memory location Loc 2
Add R1, R2, R1; Add R1 and R2 and save result in R1
Dec R2;         Decrement R2
Dec R1;         Decrement R1
Mpy R1, R2, R3; Multiply R1 and R2 and save result in R3
Store R3, Loc 3; Store R3 in memory location Loc 3
```

What is the number of cycles needed to execute the above code segment assuming each instruction takes one cycle to execute?

A. 7 B. 10 C. 13 D. 14

gateit-2007 co-and-architecture machine-instruction normal

Answer key

1.21.21 Machine Instruction: GATE IT 2008 | Question: 38

Assume that EA = (X)+ is the effective address equal to the contents of location X, with X incremented by one word length after the effective address is calculated; EA = –(X) is the effective address equal to the contents of location X, with X decremented by one word length before the effective address is calculated; EA = (X)– is the effective address equal to the contents of location X, with X decremented by one word length after the effective address is calculated. The format of the instruction is (opcode, source, destination), which means (destination ← source op destination). Using X as a stack pointer, which of the following instructions can pop the top two elements from the stack, perform the addition operation and push the result back to the stack.

A. ADD (X)–, (X)
B. ADD (X), (X)–
C. ADD –(X), (X)+
D. ADD –(X), (X)

gateit-2008 co-and-architecture machine-instruction normal

Answer key

1.22 Memory Interfacing (6)

Practice Test: Test 1 (10Q)

1.22.1 Memory Interfacing: GATE CSE 1990 | Question: 4-iv

State whether the following statements are TRUE or FALSE with reason:

Transferring data in blocks from the main memory to the cache memory enables an interleaved main memory unit to operate at its maximum speed.

gate1990 true-false co-and-architecture cache-memory memory-interfacing

Answer key

1.22.2 Memory Interfacing: GATE CSE 1991 | Question: 01,ii

In interleaved memory organization, consecutive words are stored in consecutive memory modules in ________ interleaving, whereas consecutive words are stored within the module in ________ interleaving.

gate1991 co-and-architecture normal memory-interfacing descriptive

Answer key

1.22.3 Memory Interfacing: GATE CSE 2006 | Question: 41

A CPU has a cache with block size 64 bytes. The main memory has k banks, each bank being c bytes wide. Consecutive c – byte chunks are mapped on consecutive banks with wrap-around. All the k banks can be accessed in parallel, but two accesses to the same bank must be serialized. A cache block access may involve multiple iterations of parallel bank accesses depending on the amount of data obtained by accessing all the k banks in parallel. Each iteration requires decoding the bank numbers to be accessed in parallel and this takes $\frac{k}{2}ns$. The latency of one bank access is 80 ns. If $c=2$ and $k=24$, the latency of retrieving a cache block starting at address zero from main memory is:

A. 92 ns B. 104 ns C. 172 ns D. 184 ns

gatecse-2006 co-and-architecture cache-memory memory-interfacing normal

Answer key

1.22.4 Memory Interfacing: GATE CSE 2016 | Set 1 | Question: 09

A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of address bus of the processor is at least __________ bits.

gatecse-2016-set1 co-and-architecture easy numerical-answers memory-interfacing

Answer key

1.22.5 Memory Interfacing: GATE CSE 2018 | Question: 23

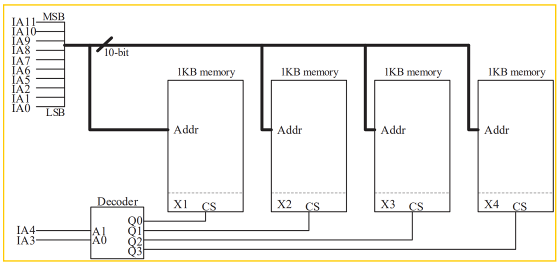
A 32-bit wide main memory unit with a capacity of 1 GB is built using $256M \times 4$-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is __________.

gatecse-2018 co-and-architecture memory-interfacing normal numerical-answers one-mark

Answer key

1.22.6 Memory Interfacing: GATE CSE 2023 | Question: 32

A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11-IA0 are connected to the address port of these blocks. The chip select (CS) is active high.



The input memory addresses (IA11-IA0), in decimal, for the starting locations (Addr = 0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

A. $(0,1,2,3)$ B. $(0,1024,2048,3072)$ C. $(0,8,16,24)$ D. $(0,0,0,0)$

gatecse-2023 co-and-architecture memory-interfacing two-marks

Answer key

1.23 Microprogramming (12)

Practice Test: Test 1 (15Q)

1.23.1 Microprogramming: GATE CSE 1987 | Question: 4a

Find out the width of the control memory of a horizontal microprogrammed control unit, given the following specifications:

- 16 control lines for the processor consisting of ALU and 7 registers.
- Conditional branching facility by checking 4 status bits.
- Provision to hold 128 words in the control memory.

gate1987 co-and-architecture microprogramming descriptive

Answer key

1.23.2 Microprogramming: GATE CSE 1996 | Question: 2.25

A micro program control unit is required to generate a total of 25 control signals. Assume that during any micro instruction, at most two control signals are active. Minimum number of bits required in the control word to generate the required control signals will be:

A. 2 B. 2.5 C. 10 D. 12

gate1996 co-and-architecture microprogramming normal

Answer key

1.23.3 Microprogramming: GATE CSE 1997 | Question: 5.3

A micro instruction is to be designed to specify:

a. none or one of the three micro operations of one kind and
b. none or upto six micro operations of another kind

The minimum number of bits in the micro-instruction is:

A. 9 B. 5 C. 8 D. None of the above

gate1997 co-and-architecture microprogramming normal

Answer key

1.23.4 Microprogramming: GATE CSE 1999 | Question: 2.19

Arrange the following configuration for CPU in decreasing order of operating speeds:
Hard wired control, Vertical microprogramming, Horizontal microprogramming.

A. Hard wired control, Vertical microprogramming, Horizontal microprogramming.
B. Hard wired control, Horizontal microprogramming, Vertical microprogramming.
C. Horizontal microprogramming, Vertical microprogramming, Hard wired control.
D. Vertical microprogramming, Horizontal microprogramming, Hard wired control.

gate1999 co-and-architecture microprogramming normal

Answer key

1.23.5 Microprogramming: GATE CSE 2002 | Question: 2.7

Horizontal microprogramming:

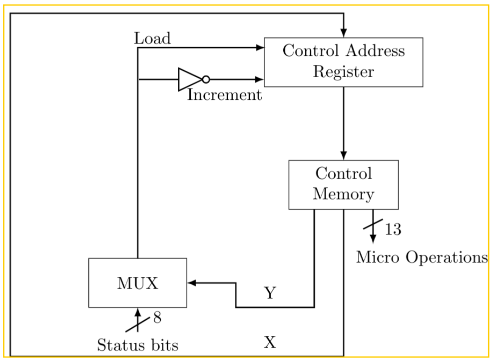
A. does not require use of signal decoders
B. results in larger sized microinstructions than vertical microprogramming
C. uses one bit for each control signal
D. all of the above

gatecse-2002 co-and-architecture microprogramming

Answer key

1.23.6 Microprogramming: GATE CSE 2004 | Question: 67

The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the input of the MUX.



How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

A. $10, 3, 1024$ B. $8, 5, 256$ C. $5, 8, 2048$ D. $10, 3, 512$

gatecse-2004 co-and-architecture microprogramming normal

Answer key

1.23.7 Microprogramming: GATE CSE 2013 | Question: 28

Consider the following sequence of micro-operations.

```
MBR ← PC
MAR ← X  PC ← Y
Memory ← MBR
```

Which one of the following is a possible operation performed by this sequence?

A. Instruction fetch
B. Operand fetch
C. Conditional branch
D. Initiation of interrupt service

gatecse-2013 co-and-architecture microprogramming normal

Answer key

1.23.8 Microprogramming: GATE IT 2004 | Question: 49

A CPU has only three instructions $I1, I2$ and $I3$, which use the following signals in time steps $T1 - T5$:

$I1 : T1$: Ain, Bout, Cin
$T2$: PCout, Bin
$T3$: Zout, Ain
$T4$: Bin, Cout
$T5$: End

$I2 : T1$: Cin, Bout, Din
$T2$: Aout, Bin
$T3$: Zout, Ain
$T4$: Bin, Cout
$T5$: End

$I3 : T1$: Din, Aout
$T2$: Ain, Bout
$T3$: Zout, Ain
$T4$: Dout, Ain
$T5$: End

Which of the following logic functions will generate the hardwired control for the signal Ain ?

A. $T1.I1 + T2.I3 + T4.I3 + T3$
B. $(T1 + T2 + T3).I3 + T1.I1$
C. $(T1 + T2).I1 + (T2 + T4).I3 + T3$
D. $(T1 + T2).I2 + (T1 + T3).I1 + T3$

gateit-2004 co-and-architecture microprogramming normal

Answer key

1.23.9 Microprogramming: GATE IT 2005 | Question: 45

A hardwired CPU uses 10 control signals S_1 to S_{10}, in various time steps T_1 to T_5, to implement 4 instructions I_1 to I_4 as shown below:

	$\mathbf{T_1}$	$\mathbf{T_2}$	$\mathbf{T_3}$	$\mathbf{T_4}$	$\mathbf{T_5}$
$\mathbf{I_1}$	S_1, S_3, S_5	S_2, S_4, S_6	S_1, S_7	S_{10}	S_3, S_8
$\mathbf{I_2}$	S_1, S_3, S_5	S_8, S_9, S_{10}	$S_5, S_6 S_7$	S_6	S_{10}
$\mathbf{I_3}$	S_1, S_3, S_5	S_7, S_8, S_{10}	S_2, S_6, S_9	S_{10}	S_1, S_3
$\mathbf{I_4}$	S_1, S_3, S_5	S_2, S_6, S_7	S_5, S_{10}	S_6, S_9	S_{10}

Which of the following pairs of expressions represent the circuit for generating control signals S_5 and S_{10} respectively?

$((I_j + I_k)T_n$ indicates that the control signal should be generated in time step T_n if the instruction being executed is I_j or l_k)

A. $S_5 = T_1 + I_2 \cdot T_3$ and

$S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

B. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

C. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_2 + I_3 + I_4) \cdot T_2 + (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

D. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and

$S_{10} = (I_2 + I_3) \cdot T_2 + I_4 \cdot T_3 + (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

gateit-2005 co-and-architecture microprogramming normal

Answer key

1.23.10 Microprogramming: GATE IT 2005 | Question: 49

An instruction set of a processor has 125 signals which can be divided into 5 groups of mutually exclusive signals as follows:

Group $1 : 20$ signals, Group $2 : 70$ signals, Group $3 : 2$ signals, Group $4 : 10$ signals, Group $5 : 23$ signals.

How many bits of the control words can be saved by using vertical microprogramming over horizontal microprogramming?

A. 0 B. 103 C. 22 D. 55

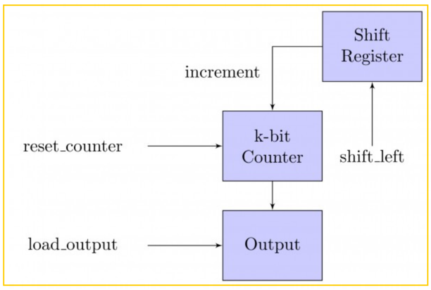
gateit-2005 co-and-architecture microprogramming normal

Answer key

1.23.11 Microprogramming: GATE IT 2006 | Question: 41

The data path shown in the figure computes the number of $1s$ in the $32-bit$ input word corresponding to an unsigned even integer stored in the shift register.

The unsigned counter, initially zero, is incremented if the most significant bit of the shift register is 1.



The microprogram for the control is shown in the table below with missing control words for microinstructions $I_1, I_2, \dots I_n$.

Microinstruction	Reset_Counter	Shift_left	Load_output
BEGIN	1	0	0
I1	?	?	?
:	:	:	:
In	?	?	?
END	0	0	1

The counter width (k), the number of missing microinstructions (n), and the control word for microinstructions $I_1, I_2, \dots I_n$ are, respectively,

A. $32, 5, 010$ B. $5, 32, 010$ C. $5, 31, 011$ D. $5, 31, 010$

gateit-2006 co-and-architecture microprogramming normal

Answer key

1.23.12 Microprogramming: GATE IT 2008 | Question: 39

Consider a CPU where all the instructions require 7 clock cycles to complete execution. There are 140 instructions in the instruction set. It is found that 125 control signals are needed to be generated by the control unit. While designing the horizontal microprogrammed control unit, single address field format is used for branch control logic. What is the minimum size of the control word and control address register?

A. $125, 7$ B. $125, 10$ C. $135, 9$ D. $135, 10$

gateit-2008 co-and-architecture microprogramming normal

Answer key

1.24 Pipelining (39)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (10Q)

1.24.1 Pipelining: GATE CSE 1999 | Question: 13

An instruction pipeline consists of 4 stages – Fetch (F), Decode field (D), Execute (E) and Result Write (W). The 5 instructions in a certain instruction sequence need these stages for the different number of clock cycles as shown by the table below

Instruction	F	D	E	W
1	1	2	1	1
2	1	2	2	1
3	2	1	3	2
4	1	3	2	1
5	1	2	1	2

Find the number of clock cycles needed to perform the 5 instructions.

gate1999 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.2 Pipelining: GATE CSE 2000 | Question: 1.8

Comparing the time T1 taken for a single instruction on a pipelined CPU with time T2 taken on a non-pipelined but identical CPU, we can say that

A. T1 ≤ T2
B. T1 ≥ T2
C. T1 < T2
D. T1 and T2 plus the time taken for one instruction fetch cycle

gatecse-2000 pipelining co-and-architecture easy

Answer key

1.24.3 Pipelining: GATE CSE 2000 | Question: 12

An instruction pipeline has five stages where each stage take 2 nanoseconds and all instruction use all five stages. Branch instructions are not overlapped. i.e., the instruction after the branch is not fetched till the branch instruction is completed. Under ideal conditions,

A. Calculate the average instruction execution time assuming that 20% of all instructions executed are branch instruction. Ignore the fact that some branch instructions may be conditional.
B. If a branch instruction is a conditional branch instruction, the branch need not be taken. If the branch is not taken, the following instructions can be overlapped. When 80% of all branch instructions are conditional branch instructions, and 50% of the conditional branch instructions are such that the branch is taken, calculate the average instruction execution time.

gatecse-2000 co-and-architecture pipelining normal descriptive

Answer key

1.24.4 Pipelining: GATE CSE 2001 | Question: 12

Consider a $5-$stage pipeline - IF (Instruction Fetch), ID (Instruction Decode and register read), EX (Execute), MEM (memory), and WB (Write Back). All (memory or register) reads take place in the second phase of a clock cycle and all writes occur in the first phase. Consider the execution of the following instruction sequence:

I1:	sub $r2, r3, r4$	/* $r2 \leftarrow r3 - r4$ */
I2:	sub $r4, r2, r3$	/* $r4 \leftarrow r2 - r3$ */
I3:	sw $r2, 100(r1)$	/* $M[r1 + 100] \leftarrow$ r2 */
I4:	sub $r3, r4, r2$	/* $r3 \leftarrow r4 - r2$ */

A. Show all data dependencies between the four instructions.
B. Identify the data hazards.
C. Can all hazards be avoided by forwarding in this case.

gatecse-2001 co-and-architecture pipelining normal descriptive

Answer key

1.24.5 Pipelining: GATE CSE 2002 | Question: 2.6, ISRO2008-19

The performance of a pipelined processor suffers if:

A. the pipeline stages have different delays
B. consecutive instructions are dependent on each other
C. the pipeline stages share hardware resources
D. All of the above

gatecse-2002 co-and-architecture pipelining easy isro2008

Answer key

1.24.6 Pipelining: GATE CSE 2003 | Question: 10, ISRO-DEC2017-41

For a pipelined CPU with a single ALU, consider the following situations

I. The $j+1^{st}$ instruction uses the result of the j^{th} instruction as an operand
II. The execution of a conditional jump instruction
III. The j^{th} and $j+1^{st}$ instructions require the ALU at the same time.

Which of the above can cause a hazard

A. I and II only
B. II and III only
C. III only
D. All the three

gatecse-2003 co-and-architecture pipelining normal isrodec2017

Answer key

1.24.7 Pipelining: GATE CSE 2004 | Question: 69

A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 $nanoseconds$, respectively. Registers that are used between the stages have a delay of 5 $nanoseconds$ each. Assuming constant clocking rate, the total time taken to process 1000 data items on this pipeline will be:

A. 120.4 microseconds
B. 160.5 microseconds
C. 165.5 microseconds
D. 590.0 microseconds

gatecse-2004 co-and-architecture pipelining normal

Answer key

1.24.8 Pipelining: GATE CSE 2006 | Question: 42

A CPU has a five-stage pipeline and runs at 1 GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline. The processor stops fetching new instructions following a conditional branch until the branch outcome is known. A program executes 10^9 instructions out of which 20% are conditional branches. If each instruction takes one cycle to complete on average, the total execution time of the program is:

A. 1.0 second
B. 1.2 seconds
C. 1.4 seconds
D. 1.6 seconds

gatecse-2006 co-and-architecture pipelining normal

Answer key

1.24.9 Pipelining: GATE CSE 2007 | Question: 37, ISRO2009-37

Consider a pipelined processor with the following four stages:

- IF: Instruction Fetch
- ID: Instruction Decode and Operand Fetch
- EX: Execute
- WB: Write Back

The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD	R2, R1, R0	$R2 \leftarrow R1+R0$
MUL	R4, R3, R2	$R4 \leftarrow R3*R2$
SUB	R6, R5, R4	$R6 \leftarrow R5-R4$

A. 7
B. 8
C. 10
D. 14

gatecse-2007 co-and-architecture pipelining normal isro2009

Answer key

1.24.10 Pipelining: GATE CSE 2008 | Question: 76

Delayed branching can help in the handling of control hazards

For all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false,

A. The instruction following the conditional branch instruction in memory is executed
B. The first instruction in the fall through path is executed
C. The first instruction in the taken path is executed
D. The branch takes longer to execute than any other instruction

gatecse-2008 co-and-architecture pipelining normal

Answer key

1.24.11 Pipelining: GATE CSE 2008 | Question: 77

Delayed branching can help in the handling of control hazards

The following code is to run on a pipelined processor with one branch delay slot:

I1: ADD $R2 \leftarrow R7 + R8$

I2: Sub $R4 \leftarrow R5 - R6$

I3: ADD $R1 \leftarrow R2 + R3$

I4: STORE Memory $[R4] \leftarrow R1$

BRANCH to Label if $R1 == 0$

Which of the instructions I1, I2, I3 or I4 can legitimately occupy the delay slot without any program modification?

A. I1 B. I2 C. I3 D. I4

gatecse-2008 co-and-architecture pipelining normal

Answer key

1.24.12 Pipelining: GATE CSE 2009 | Question: 28

Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions $I1, I2, I3, I4$ in stages $S1, S2, S3, S4$ is shown below:

	S1	S2	S3	S4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop?

For (i=1 to 2) {I1; I2; I3; I4;}

A. 16 B. 23 C. 28 D. 30

gatecse-2009 co-and-architecture pipelining normal

Answer key

1.24.13 Pipelining: GATE CSE 2010 | Question: 33

A $5-$stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, OF and WO stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?

Instruction	Meaning of instruction
t_0: MUL R_2, R_0, R_1	$R_2 \leftarrow R_0 * R_1$
t_1: DIV R_5, R_3, R_4	$R_5 \leftarrow R_3 / R_4$
t_2: ADD R_2, R_5, R_2	$R_2 \leftarrow R_5 + R_2$
t_3 : SUB R_5, R_2, R_6	$R_5 \leftarrow R_2 - R_6$

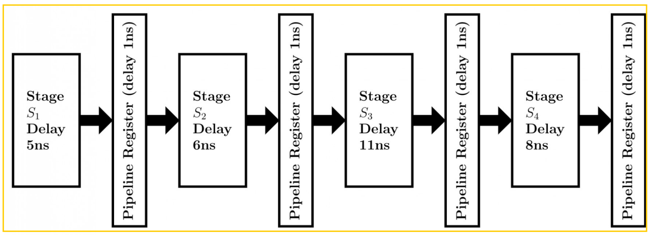
A. 13 B. 15 C. 17 D. 19

gatecse-2010 co-and-architecture pipelining normal

Answer key

1.24.14 Pipelining: GATE CSE 2011 | Question: 41

Consider an instruction pipeline with four stages (S1, S2, S3 and S4) each with combinational circuit only. The pipeline registers are required between each stage and at the end of the last stage. Delays for the stages and for the pipeline registers are as given in the figure.



What is the approximate speed up of the pipeline in steady state under ideal conditions when compared to the corresponding non-pipeline implementation?

A. 4.0 B. 2.5 C. 1.1 D. 3.0

gatecse-2011 co-and-architecture pipelining normal

Answer key

1.24.15 Pipelining: GATE CSE 2012 | Question: 20, ISRO2016-23

Register renaming is done in pipelined processors:

A. as an alternative to register allocation at compile time
B. for efficient access to function parameters and local variables
C. to handle certain kinds of hazards
D. as part of address translation

gatecse-2012 co-and-architecture pipelining easy isro2016

Answer key

1.24.16 Pipelining: GATE CSE 2013 | Question: 45

Consider an instruction pipeline with five stages without any branch prediction:

Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions $I1, I2, I3, \ldots, I12$ is executed in this pipelined processor. Instruction $I4$ is the only branch instruction and its branch target is $I9$. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

A. 132 B. 165
C. 176 D. 328

gatecse-2013 normal co-and-architecture pipelining

Answer key

1.24.17 Pipelining: GATE CSE 2014 | Set 3 | Question: 43

An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (EX), memory access (MEM), and register writeback (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively (ns stands for nanoseconds). To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency $2.2/3$ ns. Also, the EX stage is split into two stages (EX1, EX2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the EX stage and produce the next instruction pointer at the end of the EX stage in the old design and at the end of the EX2 stage in the new design. The IF stage stalls after fetching a branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is __________.

gatecse-2014-set3 co-and-architecture pipelining numerical-answers normal

Answer key

1.24.18 Pipelining: GATE CSE 2014 | Set 3 | Question: 9

Consider the following processors (ns stands for nanoseconds). Assume that the pipeline registers have zero latency.

- P1: Four-stage pipeline with stage latencies 1 ns, 2 ns, 2 ns, 1 ns.
- P2: Four-stage pipeline with stage latencies 1 ns, 1.5 ns, 1.5 ns, 1.5 ns.
- P3: Five-stage pipeline with stage latencies 0.5 ns, 1 ns, 1 ns, 0.6 ns, 1 ns.
- P4: Five-stage pipeline with stage latencies 0.5 ns, 0.5 ns, 1 ns, 1 ns, 1.1 ns.

Which processor has the highest peak clock frequency?

A. P1 B. P2 C. P3 D. P4

gatecse-2014-set3 co-and-architecture pipelining normal

Answer key

1.24.19 Pipelining: GATE CSE 2015 | Set 1 | Question: 38

Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is _______________.

gatecse-2015-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.20 Pipelining: GATE CSE 2015 | Set 2 | Question: 44

Consider the sequence of machine instruction given below:

$$\begin{array}{ll} \text{MUL} & \text{R5, R0, R1} \\ \text{DIV} & \text{R6, R2, R3} \\ \text{ADD} & \text{R7, R5, R6} \\ \text{SUB} & \text{R8, R7, R4} \end{array}$$

In the above sequence, $R0$ to $R8$ are general purpose registers. In the instructions shown, the first register shows the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following 4 stages: (1) Instruction Fetch and Decode (IF), (2) Operand Fetch (OF), (3) Perform Operation (PO) and (4) Write back the result (WB). The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instruction is __________.

gatecse-2015-set2 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.21 Pipelining: GATE CSE 2015 | Set 3 | Question: 51

Consider the following reservation table for a pipeline having three stages S_1, S_2 and S_3.

Time $\rightarrow$					
	1	2	3	4	5
S_1	X				X
S_2		X		X	
S_3			X		

The minimum average latency (MAL) is ______

gatecse-2015-set3 co-and-architecture pipelining difficult numerical-answers

Answer key

1.24.22 Pipelining: GATE CSE 2016 | Set 1 | Question: 32

The stage delays in a 4-stage pipeline are $800, 500, 400$ and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionality equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is ____________ percent.

gatecse-2016-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.23 Pipelining: GATE CSE 2016 | Set 2 | Question: 33

Consider a 3 GHz (gigahertz) processor with a three stage pipeline and stage latencies τ_1, τ_2 and τ_3 such that $\tau_1 = \frac{3\tau_2}{4} = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency , the new frequency is ___________ GHz, ignoring delays in the pipeline registers.

gatecse-2016-set2 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.24 Pipelining: GATE CSE 2017 | Set 1 | Question: 50

Instruction execution in a processor is divided into 5 stages, *Instruction Fetch* (IF), *Instruction Decode* (ID), *Operand fetch* (OF), *Execute* (EX), and *Write Back* (WB). These stages take **5, 4, 20, 10** and **3 nanoseconds (ns)** respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of **2 ns**. Two pipelined implementation of the processor are contemplated:

i. a naive pipeline implementation (NP) with 5 stages and
ii. an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of **12 ns** and **8 ns** respectively.

The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is __________ .

gatecse-2017-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.25 Pipelining: GATE CSE 2018 | Question: 50

The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is _____.

gatecse-2018 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.26 Pipelining: GATE CSE 2020 | Question: 43

Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____________.

gatecse-2020 numerical-answers co-and-architecture pipelining two-marks

Answer key

1.24.27 Pipelining: GATE CSE 2021 | Set 1 | Question: 53

A five-stage pipeline has stage delays of $150, 120, 150, 160$ and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each.
The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is ________ nanoseconds.

gatecse-2021-set1 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.28 Pipelining: GATE CSE 2021 | Set 2 | Question: 53

Consider a pipelined processor with 5 stages, Instruction Fetch(IF), Instruction Decode(ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX stage. The EX stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies.

Consider the following sequence of 8 instructions:

ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL

Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The $speedup$ defined as follows.

$$Speedup = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forearding}}$$

The $Speedup$ achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____________

gatecse-2021-set2 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.29 Pipelining: GATE CSE 2023 | Question: 23

Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle.

The total execution time for executing 100 instructions on this processor is ____________ ns.

gatecse-2023 co-and-architecture pipelining numerical-answers one-mark

Answer key

1.24.30 Pipelining: GATE CSE 2024 | Set 1 | Question: 20

Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about *forwarding* is/are CORRECT?

A. In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction
B. In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction
C. Forwarding cannot prevent all pipeline stalls
D. Forwarding does not require any extra hardware to retrieve the data from the pipeline stages

gatecse-2024-set1 multiple-selects co-and-architecture pipelining one-mark

Answer key

1.24.31 Pipelining: GATE CSE 2024 | Set 2 | Question: 21

An instruction format has the following structure:

Instruction Number: *Opcode destination reg, source reg*-1, *source reg*-2

Consider the following sequence of instructions to be executed in a pipelined processor:

$\mathrm{I}\,1$: DIV R3, R1, R2
$\mathrm{I}\,2$: SUB R5, R3, R4
$\mathrm{I}\,3$: ADD R3, R5, R6
$\mathrm{I}\,4$: MUL R7, R3, R8

Which of the following statements is/are TRUE?

A. There is a RAW dependency on $\mathrm{R}\,3$ between $\mathrm{I}\,1$ and $\mathrm{I}\,2$
B. There is a WAR dependency on $\mathrm{R}\,3$ between $\mathrm{I}\,1$ and $\mathrm{I}\,3$
C. There is a RAW dependency on $\mathrm{R}\,3$ between $\mathrm{I}\,2$ and $\mathrm{I}\,3$
D. There is a WAW dependency on $\mathrm{R}\,3$ between $\mathrm{I}\,3$ and $\mathrm{I}\,4$

gatecse-2024-set2 co-and-architecture multiple-selects pipelining one-mark

Answer key

1.24.32 Pipelining: GATE CSE 2024 | Set 2 | Question: 48

A non-pipelined instruction execution unit operating at $2\mathrm{GHz}$ takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5 -stage pipeline at $2\mathrm{GHz}$. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (rounded off to one decimal place) obtained by the pipelined design over the non-pipelined design is ____________.

gatecse-2024-set2 numerical-answers co-and-architecture pipelining two-marks

Answer key

1.24.33 Pipelining: GATE CSE 2025 | Set 2 | Question: 46

A 5-stage instruction pipeline has stage delays of $180, 250, 150, 170,$ and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is ________. (rounded off to two decimal places)

gatecse2025-set2 co-and-architecture pipelining numerical-answers easy two-marks

Answer key

1.24.34 Pipelining: GATE IT 2004 | Question: 47

Consider a pipeline processor with 4 stages $S1$ to $S4$. We want to execute the following loop:

```
for (i = 1; i < = 1000; i++)
  {I1, I2, I3, I4}
```

where the time taken (in ns) by instructions $I1$ to $I4$ for stages $S1$ to $S4$ are given below:

	S_1	S_2	S_3	S_4
I1	1	2	1	2
I2	2	1	2	1
I3	1	1	2	1
I4	2	1	2	1

The output of $I1$ for $i = 2$ will be available after

A. 11 ns
B. 12 ns
C. 13 ns
D. 28 ns

gateit-2004 co-and-architecture pipelining normal

Answer key

1.24.35 Pipelining: GATE IT 2005 | Question: 44

We have two designs $D1$ and $D2$ for a synchronous pipeline processor. $D1$ has 5 pipeline stages with execution times of 3 nsec, 2 nsec, 4 nsec, 2 nsec and 3 nsec while the design $D2$ has 8 pipeline stages each with 2 nsec execution time How much time can be saved using design $D2$ over design $D1$ for executing 100 instructions?

A. 214 nsec
B. 202 nsec
C. 86 nsec
D. -200 nsec

gateit-2005 co-and-architecture pipelining normal

Answer key

1.24.36 Pipelining: GATE IT 2006 | Question: 78

A pipelined processor uses a $4-$stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q))/T$ is given below. The values of variables P, Q, R, S and T are available in the registers $R0, R1, R2, R3$ and $R4$ respectively, before the execution of the instruction sequence.

ADD	R5, R0, R1	; R5 ← R0 + R1
MUL	R6, R2, R5	; R6 ← R2 * R5
SUB	R5, R3, R6	; R5 ← R3 - R6
DIV	R6, R5, R4	; R6 ← R5/R4
STORE	R6, X	; X ← R6

The number of Read-After-Write (RAW) dependencies, Write-After-Read(WAR) dependencies, and Write-After-Write (WAW) dependencies in the sequence of instructions are, respectively,

A. $2,2,4$ B. $3,2,3$ C. $4,2,2$ D. $3,3,2$

gateit-2006 co-and-architecture pipelining normal

Answer key

1.24.37 Pipelining: GATE IT 2006 | Question: 79

A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q))/T$ is given below. The values of variables P, Q, R, S and T are available in the registers $R0, R1, R2, R3$ and $R4$ respectively, before the execution of the instruction sequence.

$$\begin{array}{lll}
\text{ADD} & R5, R0, R1 & ; R5 \leftarrow \text{R0} + \text{R1} \\
\text{MUL} & R6, R2, R5 & ; R6 \leftarrow \text{R2} * \text{R5} \\
\text{SUB} & R5, R3, R6 & ; R5 \leftarrow \text{R3} - \text{R6} \\
\text{DIV} & R6, R5, R4 & ; R6 \leftarrow \text{R5}/\text{R4} \\
\text{STORE} & R6, X & ; X \leftarrow \text{R6}
\end{array}$$

The IF, ID and WB stages take 1 clock cycle each. The EX stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL and DIV operations. Operand forwarding from the EX stage to the ID stage is used. The number of clock cycles required to complete the sequence of instructions is

A. 10 B. 12 C. 14 D. 16

gateit-2006 co-and-architecture pipelining normal

Answer key

1.24.38 Pipelining: GATE IT 2007 | Question: 6, ISRO2011-25

A processor takes 12 cycles to complete an instruction I. The corresponding pipelined processor uses 6 stages with the execution times of $3, 2, 5, 4, 6$ and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

A. 1.83 B. 2 C. 3 D. 6

gateit-2007 co-and-architecture pipelining normal isro2011

Answer key

1.24.39 Pipelining: GATE IT 2008 | Question: 40

A non pipelined single cycle processor operating at 100 MHz is converted into a synchronous pipelined processor with five stages requiring 2.5 nsec, 1.5 nsec, 2 nsec, 1.5 nsec and 2.5 nsec, respectively. The delay of the latches is 0.5 nsec. The speedup of the pipeline processor for a large number of instructions is:

A. 4.5 B. 4.0 C. 3.33 D. 3.0

gateit-2008 co-and-architecture pipelining normal

Answer key

1.25 Runtime Environment (2)

1.25.1 Runtime Environment: GATE CSE 2001 | Question: 1.10, UGCNET-Dec2012-III: 36

Suppose a processor does not have any stack pointer registers, which of the following statements is true?

A. It cannot have subroutine call instruction
B. It cannot have nested subroutines call
C. Interrupts are not possible
D. All subroutine calls and interrupts are possible

gatecse-2001 co-and-architecture normal ugcnetcse-dec2012-paper3 runtime-environment

Answer key

1.25.2 Runtime Environment: GATE CSE 2008 | Question: 37, ISRO2009-38

The use of multiple register windows with overlap causes a reduction in the number of memory accesses for:

I. Function locals and parameters
II. Register saves and restores
III. Instruction fetches

A. I only B. II only C. III only D. I, II and III

gatecse-2008 co-and-architecture normal isro2009 runtime-environment

Answer key

1.26 Speedup (6)

Practice Test: Test (11Q)

1.26.1 Speedup: GATE CSE 2014 | Set 1 | Question: 43

Consider a 6-stage instruction pipeline, where all stages are perfectly balanced. Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution if 25% of the instructions incur 2 pipeline stall cycles is __________________

gatecse-2014-set1 co-and-architecture pipelining numerical-answers normal speedup

Answer key

1.26.2 Speedup: GATE CSE 2014 | Set 1 | Question: 55

Consider two processors P_1 and P_2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P_2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P_1. If the clock frequency of P_1 is 1GHZ, then the clock frequency of P_2 (in GHz) is _______.

gatecse-2014-set1 co-and-architecture numerical-answers normal speedup

Answer key

1.26.3 Speedup: GATE CSE 2024 | Set 1 | Question: 45

The baseline execution time of a program on a 2GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is $10\ ns$ when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is ___________.

gatecse-2024-set1 numerical-answers co-and-architecture speedup two-marks

Answer key

1.26.4 Speedup: GATE CSE 2026 | Set 2 | Question: 47

A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is ________ (rounded off to two decimal places)

Note: $1\text{G} = 10^9$

gatecse-2026-set2 co-and-architecture pipelining speedup numerical-answers two-marks

Answer key

1.26.5 Speedup: GATE IT 2004 | Question: 50

In an enhancement of a design of a CPU, the speed of a floating point unit has been increased by 20% and the speed of a fixed point unit has been increased by 10%. What is the overall speedup achieved if the ratio of the number of floating point operations to the number of fixed point operations is $2:3$ and the floating point operation used to take twice the time taken by the fixed point operation in the original design?

A. 1.155 B. 1.185 C. 1.255 D. 1.285

gateit-2004 normal co-and-architecture speedup

Answer key

1.26.6 Speedup: GATE IT 2007 | Question: 36

The floating point unit of a processor using a design D takes $2t$ cycles compared to t cycles taken by the fixed point unit. There are two more design suggestions D_1 and D_2. D_1 uses 30% more cycles for fixed point unit but 30% less cycles for floating point unit as compared to design D. D_2 uses 40% less cycles for fixed point unit but 10% more cycles for floating point unit as compared to design D. For a given program which has 80% fixed point operations and 20% floating point operations, which of the following ordering reflects the relative performances of three designs?
($D_i > D_j$ denotes that D_i is faster than D_j)

A. $D_1 > D > D_2$
B. $D_2 > D > D_1$
C. $D > D_2 > D_1$
D. $D > D_1 > D_2$

gateit-2007 co-and-architecture normal speedup

Answer key

1.27 Stall (1)

1.27.1 Stall: GATE CSE 2022 | Question: 51

A processor X_1 operating at 2 GHz has a standard $5-$stage RISC instruction pipeline having a base $\text{CPI (cycles per instruction)}$ of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2. If the BPU has a prediction accuracy of 80%, the speed up $\textit{(rounded off to two decimal places)}$ obtained by X_2 over X_1 in executing P is ________________.

gatecse-2022 numerical-answers co-and-architecture pipelining stall two-marks

Answer key

1.28 Virtual Memory (3)

1.28.1 Virtual Memory: GATE CSE 1991 | Question: 03,iii

The total size of address space in a virtual memory system is limited by:

A. the length of MAR
B. the available secondary storage
C. the available main memory
D. all of the above
E. none of the above

gate1991 co-and-architecture virtual-memory normal multiple-selects

Answer key

1.28.2 Virtual Memory: GATE CSE 2004 | Question: 47

Consider a system with a two-level paging scheme in which a regular memory access takes 150 $\textit{nanoseconds}$, and servicing a page fault takes 8 $\textit{milliseconds}$. An average instruction takes 100 nanoseconds of CPU time, and two memory accesses. The TLB hit ratio is 90%, and the page fault rate is one in every $10,000$ instructions. What is the effective average instruction execution time?

A. 645 nanoseconds
B. 1050 nanoseconds
C. 1215 nanoseconds
D. 1230 nanoseconds

gatecse-2004 co-and-architecture virtual-memory normal

Answer key

1.28.3 Virtual Memory: GATE CSE 2008 | Question: 38

In an instruction execution pipeline, the earliest that the data TLB (Translation Lookaside Buffer) can be accessed is:

A. before effective address calculation has started

B. during effective address calculation

C. after effective address calculation has completed

D. after data cache lookup has completed

gatecse-2008 co-and-architecture virtual-memory normal

Answer key

Answer Keys

1.1.1	C	1.1.2	N/A	1.1.3	N/A	1.1.4	N/A	1.1.5	B
1.1.6	D	1.1.7	A;B;C;D	1.1.8	C	1.1.9	A	1.1.10	B
1.1.11	C	1.1.12	B	1.1.13	C	1.1.14	C	1.1.15	D
1.1.16	D	1.1.17	B	1.1.18	D	1.1.19	D	1.2.1	180
1.2.2	11.83:11.87	1.2.3	4:4	1.3.1	16417	1.4.1	A;B;C;D	1.4.2	D
1.5.1	N/A	1.5.2	N/A	1.5.3	22	1.5.4	N/A	1.5.5	D
1.5.6	D	1.5.7	61.25	1.5.8	N/A	1.5.9	B	1.5.10	B
1.5.11	N/A	1.5.12	N/A	1.5.13	C	1.5.14	A	1.5.15	A
1.5.16	D	1.5.17	C	1.5.18	B	1.5.19	D	1.5.20	C
1.5.21	A	1.5.22	A	1.5.23	D	1.5.24	B	1.5.25	C
1.5.26	D	1.5.27	C	1.5.28	C	1.5.29	D	1.5.30	C
1.5.31	A	1.5.32	A	1.5.33	A	1.5.34	D	1.5.35	D
1.5.36	20	1.5.37	1.68	1.5.38	14	1.5.39	A	1.5.40	24
1.5.41	30	1.5.42	0.05	1.5.43	14	1.5.44	A	1.5.45	4.7 : 4.8
1.5.46	18	1.5.47	B	1.5.48	D	1.5.49	160	1.5.50	13.3:13.3;13.5:13.5
1.5.51	B	1.5.52	17 : 17	1.5.53	2 : 2	1.5.54	A	1.5.55	A;B;D
1.5.56	0.85	1.5.57	19	1.5.58	B;C	1.5.59	3	1.5.60	A
1.5.61	B;C	1.5.62	C	1.5.63	B	1.5.64	C	1.5.65	C
1.5.66	A	1.5.67	B	1.5.68	D	1.5.69	A	1.6.1	76
1.7.1	B	1.8.1	D	1.8.2	B	1.8.3	456	1.8.4	80000 : 80000
1.8.5	A	1.8.6	C	1.8.7	C	1.8.8	C	1.9.1	A
1.10.1	A	1.10.2	B	1.10.3	B	1.10.4	B	1.11.1	B
1.12.1	N/A	1.12.2	D	1.12.3	B	1.12.4	B	1.12.5	28
1.12.6	C	1.12.7	A;B;C	1.13.1	A;B;D	1.13.2	A	1.13.3	A
1.13.4	A;B	1.13.5	128	1.14.1	77.3 : 77.3	1.15.1	95 : 95	1.16.1	True

1.16.2	True	1.16.3	False	1.16.4	A	1.16.5	1.4 : 1.5	1.16.6	B
1.16.7	B	1.16.8	A	1.17.1	False	1.17.2	N/A	1.17.3	C
1.17.4	C	1.17.5	A	1.17.6	-16.0	1.17.7	3.0:3.0	1.18.1	N/A
1.18.2	256	1.18.3	True	1.18.4	16383	1.18.5	500	1.18.6	32
1.18.7	14	1.18.8	32	1.18.9	34	1.18.10	D	1.18.11	B
1.19.1	D	1.20.1	D	1.20.2	B	1.20.3	A	1.20.4	A
1.20.5	B	1.20.6	C	1.20.7	C	1.20.8	10.2	1.20.9	A
1.20.10	C	1.21.1	N/A	1.21.2	N/A	1.21.3	N/A	1.21.4	B
1.21.5	A	1.21.6	D	1.21.7	B	1.21.8	C	1.21.9	B
1.21.10	D	1.21.11	A	1.21.12	C	1.21.13	D	1.21.14	D
1.21.15	16	1.21.16	50 : 50	1.21.17	B	1.21.18	B	1.21.19	D
1.21.20	C	1.21.21	A	1.22.1	TBA	1.22.2	N/A	1.22.3	D
1.22.4	31	1.22.5	59 : 60	1.22.6	C	1.23.1	N/A	1.23.2	C
1.23.3	C	1.23.4	B	1.23.5	D	1.23.6	A	1.23.7	D
1.23.8	A	1.23.9	D	1.23.10	B	1.23.11	D	1.23.12	D
1.24.1	15	1.24.2	B	1.24.3	N/A	1.24.4	N/A	1.24.5	D
1.24.6	D	1.24.7	C	1.24.8	C	1.24.9	B	1.24.10	A
1.24.11	D	1.24.12	B	1.24.13	B	1.24.14	B	1.24.15	C
1.24.16	B	1.24.17	1.50 : 1.60	1.24.18	C	1.24.19	3.2	1.24.20	13
1.24.21	3	1.24.22	33.0 : 34.0	1.24.23	4	1.24.24	1.50 : 1.51	1.24.25	219
1.24.26	2.15:2.18	1.24.27	17160 : 17160	1.24.28	1.87 : 1.88	1.24.29	2040	1.24.30	A;B;C
1.24.31	A	1.24.32	3	1.24.33	260.20:261.20	1.24.34	C	1.24.35	B
1.24.36	C	1.24.37	B	1.24.38	B	1.24.39	C	1.25.1	X
1.25.2	A	1.26.1	4	1.26.2	1.6	1.26.3	3	1.26.4	2.30:2.40
1.26.5	A	1.26.6	B	1.27.1	1.42:1.45	1.28.1	A;B	1.28.2	D
1.28.3	C								

Concept of layering.OSI and TCP/IP Protocol Stacks; Basics of packet, circuit and virtual circuit-switching; Data link layer: framing, error detection, Medium Access Control, Ethernet bridging; Routing protocols: shortest path, flooding, distance vector and link state routing; Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT); Transport layer: flow control and congestion control, UDP, TCP, sockets; Application layer protocols: DNS, SMTP, HTTP, FTP, **Email**.

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Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	3	2	4	3	3	2	2	1	1	1
2 Marks Count	3	3	3	1	3	3	3	4	4	3	1
Total Marks	8	9	8	6	9	9	8	10	9	7	**6**

The Computer Networks chapter in GATE Computer Science is a fundamental and high-scoring area, essential for understanding how modern computing systems communicate and interact. It covers everything from the physical transmission of data to high-level application protocols, network architecture, and security principles. Mastery of this subject is crucial not only for the GATE exam but also for a career in software development, system administration, and network engineering. Typically, Computer Networks carries a weightage of 8-12 marks in the GATE CS exam, with questions ranging from conceptual understanding of protocols and layers to numerical problems involving network performance, addressing, and error control. Question patterns often include multiple-choice questions (MCQs), multiple-select questions (MSQs), and numerical answer type (NAT) questions, testing both theoretical knowledge and problem-solving skills.

Topic-wise Key Concepts

Application Layer Protocols

These protocols are at the highest layer of the TCP/IP model, providing services directly to user applications. They define how applications on different hosts communicate and exchange data.

Key Concepts:

- **HTTP (Hypertext Transfer Protocol):** Used for web browsing, client-server model, stateless. Default port 80.
- **FTP (File Transfer Protocol):** Used for file transfer, uses two connections (control and data). Default ports 20 (data) and 21 (control).
- **DNS (Domain Name System):** Translates domain names to IP addresses. UDP port 53 for queries, TCP port 53 for zone transfers.
- **SMTP (Simple Mail Transfer Protocol):** Used for sending emails. Default port 25.
- **POP3 (Post Office Protocol v3):** Used for retrieving emails. Default port 110.
- **IMAP (Internet Message Access Protocol):** More advanced email retrieval, allows managing mail on server. Default port 143.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the functions and default port numbers of different protocols.
- **Technique:** Create a table mapping protocol to function and port number for quick recall.

ARP (Address Resolution Protocol)

ARP is a protocol used to map an IP address (Network Layer) to a physical MAC address (Data Link Layer) on a local network. It is essential for IP packets to be encapsulated into Ethernet frames for local delivery.

Key Concepts:

- **ARP Request:** Broadcasts a query to all hosts on the local network asking for the MAC address corresponding to a specific IP.
- **ARP Reply:** The host with the matching IP address sends a unicast reply containing its MAC address.
- **ARP Cache:** Hosts maintain a cache of IP-to-MAC mappings to reduce ARP traffic.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Assuming ARP works across routers; it's a local network protocol.
- **Technique:** Understand the ARP process step-by-step for tracing packet flows.

Bit Stuffing

Bit stuffing is a technique used in the Data Link Layer to prevent the flag sequence from appearing in the data portion of a frame. It ensures that the receiver correctly identifies the start and end of a frame.

Key Concepts:

- **Flag Sequence:** A unique bit pattern (e.g., 01111110) used to mark frame boundaries.
- **Stuffing Rule:** Whenever five consecutive 1s appear in the data, an extra 0 bit is inserted (stuffed) by the sender.
- **Destuffing Rule:** The receiver removes a 0 bit after five consecutive 1s.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly stuffing/destuffing bits, especially at the end of the data or near the flag sequence.
- **Technique:** Practice with examples, carefully counting consecutive 1s.

Bridges

Bridges are Data Link Layer devices that connect two or more LAN segments. They filter frames based on MAC addresses, forwarding only those frames destined for another segment, thus reducing collision domains and improving network performance.

Key Concepts:

- **MAC Address Filtering:** Bridges maintain a forwarding table (MAC address table) to decide whether to forward or filter frames.
- **Learning:** Bridges learn MAC addresses by inspecting the source MAC address of incoming frames.
- **Spanning Tree Protocol (STP):** Used to prevent loops in bridged networks by disabling redundant paths.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing bridges with routers (Layer 3) or hubs (Layer 1). Bridges operate at Layer 2.
- **Technique:** Trace frame paths in a bridged network, applying learning and forwarding rules.

CRC Polynomial (Cyclic Redundancy Check)

CRC is a powerful error detection technique used in the Data Link Layer. It appends a checksum (FCS - Frame Check Sequence) to the data, calculated using polynomial division.

Important Formulas and Results:

- Let $M(x)$ be the data polynomial and $G(x)$ be the generator polynomial of degree r. To find the CRC remainder $R(x)$:

$$\frac{x^r \cdot M(x)}{G(x)} = Q(x) + \frac{R(x)}{G(x)}$$

  where $R(x)$ is the remainder polynomial of degree less than r.
- The transmitted codeword $T(x)$ is:

$$T(x) = x^r \cdot M(x) + R(x)$$

  This means the codeword is exactly divisible by $G(x)$.

Key Properties and Identities:

- CRC can detect all single-bit errors, all double-bit errors, any odd number of errors, and all burst errors of length less than or equal to r.
- It can detect a high percentage of burst errors longer than r.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Performing binary division incorrectly (modulo-2 arithmetic, no borrows).
- **Technique:** Practice modulo-2 polynomial division. Remember to pad the data with r zeros before division.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD is a MAC protocol used in shared-medium networks like Ethernet. Stations listen before transmitting (carrier sense) and stop transmitting if a collision is detected (collision detection).

Important Formulas and Results:

- **Slot Time:** The maximum time it takes for a collision to be detected by all stations.

$$\boxed{\text{Slot Time} = 2 \times \text{Propagation Delay}}$$

- **Minimum Frame Size:** To ensure a collision is detected before the sender finishes transmitting the frame.

$$\text{Minimum Frame Size} = \text{Bandwidth} \times \text{Slot Time} = \text{Bandwidth} \times 2 \times \text{Propagation Delay}$$

Key Properties and Identities:

- Uses binary exponential backoff algorithm to resolve collisions.
- Collision domain is the segment where collisions can occur.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing propagation delay with transmission delay. Propagation delay is time for a bit to travel, transmission delay is time to put all bits on wire.
- **Technique:** Ensure units are consistent (e.g., convert Mbps to bits/second, km to meters, ms to seconds).

Channel Utilization

Channel utilization (or efficiency) measures the fraction of time the channel is actively transmitting data, rather than being idle or transmitting overhead. It's a key performance metric for network protocols.

Important Formulas and Results:

- General formula:

$$\text{Utilization} = \frac{\text{Time spent transmitting useful data}}{\text{Total time}}$$

- For Stop-and-Wait ARQ:

$$\boxed{U = \frac{1}{1+2a}}$$

  where $a = \frac{\text{Propagation Delay}}{\text{Transmission Time}} = \frac{T_p}{T_t}$. This assumes no errors and negligible ACK transmission time.
- For Sliding Window (Go-Back-N or Selective Repeat) with window size W:

$$\boxed{U = \min\left(1, \frac{W}{1+2a}\right)}$$

  This assumes no errors. If $W > 1 + 2a$, utilization can be 100%.
- For Pure Aloha:

$$S = G \cdot e^{-2G}$$

where S is throughput, G is offered load. Max $S \approx 0.184$ at $G = 0.5$.

- For Slotted Aloha:

$$S = G \cdot e^{-G}$$

Max $S \approx 0.368$ at $G = 1$.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly calculating a or not considering the full round-trip time.
- **Technique:** Clearly identify T_t and T_p from the problem statement. Remember T_p is one-way.

Communication

Communication in networks refers to the process of exchanging information between two or more entities. It involves various modes and fundamental concepts that define how data travels.

Key Concepts:

- **Simplex:** Data flows in one direction only (e.g., radio broadcast).
- **Half-Duplex:** Data flows in both directions, but not simultaneously (e.g., walkie-talkie).
- **Full-Duplex:** Data flows in both directions simultaneously (e.g., telephone call).
- **Bandwidth:** The maximum data transfer rate of a network path, typically measured in bits per second (bps).
- **Latency (Delay):** The time it takes for a data packet to travel from one point to another. Comprises transmission, propagation, processing, and queuing delays.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing bandwidth with throughput. Bandwidth is capacity, throughput is actual rate.
- **Technique:** Understand the implications of each communication mode on protocol design (e.g., collision detection in half-duplex).

Congestion Control

Congestion control mechanisms aim to prevent network collapse by regulating the rate at which senders transmit data when the network is overloaded. TCP implements several strategies for this.

Key Concepts (TCP):

- **Slow Start:** Exponentially increases congestion window (cwnd) at the beginning of a connection or after a timeout. cwnd starts at 1 MSS (Maximum Segment Size) and doubles every RTT.
- **Congestion Avoidance:** After cwnd reaches ssthresh (slow start threshold), cwnd increases linearly (by 1 MSS per RTT).
- **Fast Retransmit:** If the sender receives three duplicate ACKs, it retransmits the lost segment without waiting for a timeout.
- **Fast Recovery:** After Fast Retransmit, ssthresh is set to half of the current cwnd, and cwnd is set to ssthresh + 3 MSS. It then enters congestion avoidance.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly applying the rules for cwnd and ssthresh changes during slow start, congestion avoidance, and recovery phases.
- **Technique:** Draw a graph of cwnd vs. RTT to visualize the changes. Remember the ssthresh value is halved upon a loss.

Data Communication

Data communication encompasses the processes, technologies, and methods involved in the electronic transmission of information. It covers the physical and logical aspects of moving data between devices.

Key Concepts:

- **Signals:** Electrical or electromagnetic waves used to transmit data. Can be analog or digital.
- **Modulation:** Converting digital data into analog signals for transmission over analog media.
- **Demodulation:** Converting analog signals back into digital data.
- **Multiplexing:** Combining multiple data streams into a single stream for transmission (e.g., TDM, FDM, WDM).
- **Switching:** Techniques for connecting communication paths (e.g., circuit, packet, message switching).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing modulation with encoding, or different multiplexing techniques.
- **Technique:** Understand the purpose and basic mechanism of each concept.

Distance Vector Routing

Distance Vector Routing is a dynamic routing algorithm where each router maintains a routing table (distance vector) containing the best known distance to each destination and the next hop. Routers exchange their entire routing tables with directly connected neighbors.

Key Concepts:

- **Bellman-Ford Algorithm:** The underlying algorithm for distance vector routing.

$$D_x(y) = \min_v \{c(x, v) + D_v(y)\}$$

  where $D_x(y)$ is the cost from x to y, $c(x, v)$ is cost from x to neighbor v, and $D_v(y)$ is v's reported cost to y.
- **Count-to-Infinity Problem:** A major drawback where routing loops can cause costs to increase indefinitely.
- **Poisoned Reverse:** A mechanism to mitigate count-to-infinity by advertising infinite cost for routes learned from a neighbor back to that neighbor.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly updating routing tables or failing to identify routing loops.
- **Technique:** Simulate the routing table updates step-by-step for a given network topology and link cost changes.

Error Detection

Error detection techniques are used to determine if errors have occurred during data transmission. They add redundant bits to the data, allowing the receiver to check for integrity.

Key Concepts:

- **Parity Check:** Adds a single parity bit to make the total number of 1s even (even parity) or odd (odd parity). Detects single-bit errors.
- **Checksum:** Divides data into segments, sums them (using one's complement arithmetic), and takes the one's complement of the sum as the checksum. Detects most errors, but not as robust as CRC.
- **CRC (Cyclic Redundancy Check):** Most powerful error detection technique, uses polynomial division. (See CRC Polynomial topic).

Important Formulas and Results:

- **Internet Checksum:** Sum all 16-bit words (or specified size) of the data. If sum exceeds 16 bits, wrap around the carry. Take one's complement of the final sum.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Forgetting one's complement arithmetic for checksum calculations.
- **Technique:** Practice checksum calculations, especially with carry bits.

Ethernet

Ethernet is the most widely used LAN technology, standardized by IEEE 802.3. It defines the physical and Data Link Layer specifications for wired networks, primarily using CSMA/CD.

Key Concepts:

- **MAC Address:** A 48-bit (6-byte) unique hardware identifier assigned to each network interface card (NIC).
- **Ethernet Frame Format:** Includes Preamble, SFD (Start Frame Delimiter), Destination MAC, Source MAC, Type/Length, Data, and FCS (Frame Check Sequence/CRC).
- **Minimum/Maximum Frame Size:** Standard Ethernet has a minimum frame size of 64 bytes (including header and FCS) and a maximum of 1518 bytes.
- **CSMA/CD:** The access method used for shared Ethernet segments.

Important Formulas and Results:

- **Minimum Data Size:** 46 bytes (to ensure minimum frame size of 64 bytes, considering 18 bytes of header/trailer).

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Confusing the total frame size with the data payload size.
- **Technique:** Memorize the Ethernet frame structure and minimum/maximum sizes.

Fragmentation

Fragmentation is the process at the Network Layer (IP) where a large IP packet is divided into smaller fragments to traverse a network link with a smaller Maximum Transmission Unit (MTU).

Important Formulas and Results:

- **Offset Field:** In the IP header, indicates the position of the fragment's data relative to the original datagram's data, in units of 8 bytes.

$$\text{New Offset} = \frac{\text{Original Offset} \times 8 + \text{Data Length of Fragment}}{8}$$

- **Total Length Field:** In the IP header, indicates the total length of the current fragment (header + data).
- **More Fragments (MF) Flag:** Set to 1 for all fragments except the last one.

Key Properties and Identities:

- Fragmentation can occur at any router along the path.
- Reassembly typically occurs only at the destination host.
- The data payload of each fragment (except possibly the last) must be a multiple of 8 bytes.

Common Pitfalls and Problem-Solving Techniques:

- **Pitfall:** Incorrectly calculating the offset field or the data length of fragments, especially when the original packet's data length is not a multiple of 8.
- **Technique:** Work through fragmentation problems step-by-step, ensuring each fragment's data length is a multiple of 8 (except the last) and updating the offset correctly.

Hamming Code

Hamming code is an error-correcting code capable of detecting and correcting single-bit errors. It adds redundant parity bits at specific positions within the data.

Important Formulas and Results:

- **Number of Parity Bits (p):** To correct single-bit errors in a message of m data bits, the number of parity bits p must satisfy:

$$2^p \geq m + p + 1$$